

The asymptotic expansion and recurrence relations of several special functions in complex analysis and their applications

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Abstract

Special functions play a fundamental role across various domains of mathematics, including complex analysis, number theory, and mathematical physics. This thesis systematically investigates the asymptotic expansions and the recurrence relations of power series coefficients for several core special functions in the complex field.

First, the Maclaurin expansions for the powers of fundamental inverse trigonometric functions, such as $\arcsin^m z$ and $\arctan^m z$, are derived. By employing the equation method, governing linear difference equations are established, yielding explicit representations of the coefficients through nested summation operators.

Second, the asymptotic behaviors of the gamma function and its quotients are explored for large complex arguments. Utilizing properties of Bernoulli numbers and series transformations, several new Ramanujan-type, Burnside-type, and Windschitl-type asymptotic expansions for the gamma function are established in the complex sector $|\arg z| < \pi$. Furthermore, by applying Watson's lemma, generalized asymptotic expansions for Wallis-type and Gurland-type ratios of gamma functions are derived in the complex plane.

Finally, the thesis establishes general recurrence relations for the Maclaurin series coefficients of confluent and Gauss hypergeometric functions multiplied by algebraic factors of the form $(1 - \theta z)^p$. These general formulas are subsequently applied to generate novel, computationally efficient coefficient recurrences for a wide array of special functions, including the error function, incomplete gamma functions, modified Bessel functions, Chebyshev polynomials, and complete elliptic integrals.

Keywords: Special Functions; Asymptotic Expansions; Recurrence Relations; Gamma Function; Hypergeometric Functions; Maclaurin Series

Contents

1	Fundamental trigonometric functions and their powers' Maclaurin expansions through equation method	3
1.1	Introduction	3
1.2	Main Results	3
2	Asymptotic expansions of gamma function and its different combinations in complex fields	8
2.1	Introduction	8
2.2	Ramanujan-type, Burnside-type and Windschit-type asymptotic expansions of gamma functions	11
2.3	Asymptotic expansions for the different kinds of ratios of Gamma functions	22
3	Recurrence relations of coefficients involving hypergeometric function in complex fields	28
3.1	Introduction	28
3.2	Reccurence relations of coefficients involving confluent hypergeometric funcnion	30
3.3	Reccurence relations of coefficients involving gauss hypergeometric function	33
3.4	Applications	38
3.4.1	Applications of the confluent hypergeometric function	39
3.4.2	Applications of the Gauss hypergeometric function	42

Chapter 1

Fundamental trigonometric functions and their powers' Maclaurin expansions through equation method

1.1 Introduction

In this chapter, we mainly focus on deriving the Maclaurin expansions of some fundamental trigonometric functions' powers through the equation method, which will also be used in the later chapters.

1.2 Main Results

Proposition 1.2.1. For $z \in \mathbb{C}$ with $|z| < 1$, and for any integer $p \geq 0$, we have the formal expansion:

$$\arcsin^{2p+1} z = \sum_{n=0}^{\infty} a_{2n+1}(2p+1)z^{2n+1}, \quad (1.2.1)$$

where the coefficients are given explicitly by:

$$a_{2n+1}(2p+1) = \frac{(2n-1)!!}{(2n)!!(2n+1)}(2p+1)! \sum_{1 \leq i_1 < i_2 < \dots < i_p \leq n} \prod_{j=1}^p \frac{1}{(2i_j-1)^2}, \quad (1.2.2)$$

and the empty product for $p = 0$ is defined as 1.

Proof. Let $y_m(z) = \arcsin^m z = \sum_{n=0}^{\infty} a_n(m)z^n$. Taking the first derivative with respect to z , we have:

$$y'_m = m(\arcsin z)^{m-1} \frac{1}{\sqrt{1-z^2}} = \frac{my_{m-1}}{\sqrt{1-z^2}} \quad (1.2.3)$$

Multiplying by $\sqrt{1-z^2}$ and differentiating again yields:

$$-\frac{z}{\sqrt{1-z^2}}y'_m + \sqrt{1-z^2}y''_m = my'_{m-1} \quad (1.2.4)$$

Multiplying the entire equation by $\sqrt{1-z^2}$ and substituting $y'_{m-1} = \frac{(m-1)y_{m-2}}{\sqrt{1-z^2}}$, we obtain the differential equation:

$$(1-z^2)y''_m - zy'_m = m(m-1)y_{m-2} \quad (1.2.5)$$

Substituting the power series expansion $y_m(z) = \sum_{n=0}^{\infty} a_n(m)z^n$ into this differential equation gives:

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}(m)z^n - \sum_{n=2}^{\infty} n(n-1)a_n(m)z^n - \sum_{n=1}^{\infty} na_n(m)z^n = \sum_{n=0}^{\infty} m(m-1)a_n(m-2)z^n \quad (1.2.6)$$

Comparing the coefficients of z^n on both sides establishes the general recurrence relation for the coefficients:

$$(n+2)(n+1)a_{n+2}(m) - n^2a_n(m) = m(m-1)a_n(m-2) \quad (1.2.7)$$

For an odd power $m = 2p + 1$, the expansion contains only odd powers of z . Shifting the index by setting $n \rightarrow 2n + 1$ yields:

$$(2n+3)(2n+2)a_{2n+3}(m) - (2n+1)^2a_{2n+1}(m) = m(m-1)a_{2n+1}(m-2) \quad (1.2.8)$$

To linearize this recurrence, we factor out the base structure of the standard Taylor series ($m = 1$) by defining a sequence $C_n(m)$:

$$a_{2n+1}(m) = \frac{(2n-1)!!}{(2n)!!(2n+1)} C_n(m) \quad (1.2.9)$$

Substituting this definition into the recurrence relation and dividing the entire equation by $\frac{(2n+1)!!}{(2n)!!}$, we obtain a simplified first-order difference equation:

$$C_{n+1}(m) - C_n(m) = \frac{m(m-1)}{(2n+1)^2} C_n(m-2) \quad (1.2.10)$$

Since the expansion for $m > 1$ begins with z^m , we have $a_1(m) = 0$, which implies $C_0(m) = 0$. Summing the difference equation from $k = 0$ to $n - 1$ gives the summation operator:

$$C_n(m) = \sum_{k=0}^{n-1} \frac{m(m-1)}{(2k+1)^2} C_k(m-2) \quad (1.2.11)$$

For the base case $m = 1$ ($p = 0$), we have $C_n(1) = 1$ for all $n \geq 0$. Repeatedly applying the summation operator p times for $m = 2p + 1$ accumulates the constant multipliers as $(2p + 1)!$ and generates a p -fold nested sum:

$$\begin{aligned} C_n(2p+1) &= \sum_{k_p=0}^{n-1} \frac{(2p+1)(2p)}{(2k_p+1)^2} \left(\cdots \sum_{k_1=0}^{k_2-1} \frac{3 \cdot 2}{(2k_1+1)^2} C_{k_1}(1) \right) \\ &= (2p+1)! \sum_{0 \leq k_1 < k_2 < \cdots < k_p \leq n-1} \prod_{j=1}^p \frac{1}{(2k_j+1)^2} \end{aligned} \quad (1.2.12)$$

To express this in standard 1-based indexing, we apply the shift $i_j = k_j + 1$. The summation bounds transform to $1 \leq i_1 < i_2 < \cdots < i_p \leq n$, and the denominators become $(2i_j - 1)^2$:

$$C_n(2p+1) = (2p+1)! \sum_{1 \leq i_1 < i_2 < \cdots < i_p \leq n} \prod_{j=1}^p \frac{1}{(2i_j - 1)^2} \quad (1.2.13)$$

Substituting $C_n(2p+1)$ back into our original definition for $a_{2n+1}(2p+1)$ completes the proof. $\square$

Proposition 1.2.2. For $z \in \mathbb{C}$ with $|z| < 1$, and for any integer $p \geq 1$, we have the formal expansion:

$$\arcsin^{2p} z = \sum_{n=1}^{\infty} a_{2n}(2p) z^{2n}, \quad (1.2.14)$$

where the coefficients are given explicitly by:

$$a_{2n}(2p) = \frac{(2n-2)!!}{(2n-1)!!(2n)} (2p)! \sum_{1 \leq i_1 < i_2 < \dots < i_{p-1} \leq n-1} \prod_{j=1}^{p-1} \frac{1}{(2i_j)^2}, \quad (1.2.15)$$

and the empty product for $p = 1$ is defined as 1.

Proof. For an even power $m = 2p$, the function $\arcsin^{2p}(z)$ is an even function, implying its expansion contains only even powers of z . Shifting the index by setting $n \rightarrow 2n$ in the general recurrence relation yields:

$$(2n+2)(2n+1)a_{2n+2}(m) - (2n)^2 a_{2n}(m) = m(m-1)a_{2n}(m-2) \quad (1.2.16)$$

To linearize this recurrence, we factor out the structural components of the base expansion for $m = 2$ by defining a sequence $C_n(m)$:

$$a_{2n}(m) = \frac{(2n-2)!!}{(2n-1)!!(2n)} C_n(m) \quad (1.2.17)$$

Substituting this definition into the recurrence relation and simplifying the resulting double factorials, we obtain a first-order difference equation that is remarkably symmetric to the odd case:

$$C_{n+1}(m) - C_n(m) = \frac{m(m-1)}{(2n)^2} C_n(m-2) \quad (1.2.18)$$

Since the Maclaurin series for $\arcsin^m(z)$ starts at z^m , for $m > 2$ we have $a_2(m) = 0$, which implies $C_1(m) = 0$. Summing the difference equation from $k = 1$ to $n - 1$ gives the summation operator:

$$C_n(m) = \sum_{k=1}^{n-1} \frac{m(m-1)}{(2k)^2} C_k(m-2) \quad (1.2.19)$$

For the base case $m = 2$ ($p = 1$), the known expansion for $\arcsin^2(z)$ dictates that $C_n(2) = 2$ for all $n \geq 1$. Repeatedly applying the summation operator $p - 1$ times for $m = 2p$ accumulates the constant multipliers to $(2p)!$ and generates a $(p - 1)$ -fold nested sum:

$$\begin{aligned} C_n(2p) &= \sum_{k_{p-1}=1}^{n-1} \frac{(2p)(2p-1)}{(2k_{p-1})^2} \left(\dots \sum_{k_1=1}^{k_{2-1}} \frac{4 \cdot 3}{(2k_1)^2} C_{k_1}(2) \right) \\ &= (2p)! \sum_{1 \leq k_1 < k_2 < \dots < k_{p-1} \leq n-1} \prod_{j=1}^{p-1} \frac{1}{(2k_j)^2} \end{aligned} \quad (1.2.20)$$

Letting $i_j = k_j$, we substitute $C_n(2p)$ back into our original definition for $a_{2n}(2p)$, which completes the proof. Note that if $n < p$, the upper bound of the summation is smaller than the required indices, naturally evaluating to 0 as expected. $\square$

Remark 1.2.1. The Maclaurin series coefficients for the inverse hyperbolic power $\operatorname{arcsinh}^m(z)$ can be obtained directly from those of $\arcsin^m(z)$ without solving the associated recurrence relations anew. By utilizing the complex identity $\operatorname{arcsinh}(z) = -i \arcsin(iz)$, we quickly get its expansion

Proposition 1.2.3. For $z \in \mathbb{C}$ with $|z| < 1$, and for any integer $m \geq 1$, the Maclaurin expansion of $\arctan^m(z)$ is given by:

$$\arctan^m(z) = \sum_{N \equiv m \pmod{2}} (-1)^{\frac{N-m}{2}} A_N(m) z^N, \quad (1.2.21)$$

where $A_N(m)$ is generated by iterating the following summation operators from the base sequence $A_{2n+1}(1) = \frac{1}{2n+1}$ and $A_{2n}(1) = 0$:

$$A_{2n}(m) = \frac{m}{2n} \sum_{k=1}^n A_{2k-1}(m-1) \quad (\text{for even } m) \quad (1.2.22)$$

$$A_{2n+1}(m) = \frac{m}{2n+1} \sum_{k=0}^n A_{2k}(m-1) \quad (\text{for odd } m) \quad (1.2.23)$$

Proof. Let $y_m(z) = \arctan^m(z) = \sum_{n=0}^{\infty} a_n(m) z^n$. Taking the first derivative with respect to z gives $y'_m = m \arctan^{m-1}(z) \frac{1}{1+z^2}$, which rearranges into the first-order linear differential equation:

$$(1+z^2)y'_m = my_{m-1} \quad (1.2.24)$$

Substituting the power series expansion yields:

$$(1+z^2) \sum_{n=1}^{\infty} n a_n(m) z^{n-1} = m \sum_{n=0}^{\infty} a_n(m-1) z^n \quad (1.2.25)$$

Distributing and shifting the indices to align the powers of z^n gives:

$$\sum_{n=0}^{\infty} (n+1) a_{n+1}(m) z^n + \sum_{n=2}^{\infty} (n-1) a_{n-1}(m) z^n = \sum_{n=0}^{\infty} m a_n(m-1) z^n \quad (1.2.26)$$

Equating the coefficients of z^n for $n \geq 1$ establishes the general recurrence relation:

$$(n+1) a_{n+1}(m) + (n-1) a_{n-1}(m) = m a_n(m-1) \quad (1.2.27)$$

Because $\arctan(-z) = -\arctan(z)$, the function $y_m(z)$ shares the parity of m . Thus, $a_N(m) = 0$ for all $N \not\equiv m \pmod{2}$. To linearize the alternating signs of the expansion, we define a non-negative sequence $A_N(m)$ such that $a_N(m) = (-1)^{\frac{N-m}{2}} A_N(m)$. Substituting this definition into the recurrence relation gives:

$$(n+1)(-1)^{\frac{n+1-m}{2}} A_{n+1}(m) + (n-1)(-1)^{\frac{n-1-m}{2}} A_{n-1}(m) = m(-1)^{\frac{n-(m-1)}{2}} A_n(m-1) \quad (1.2.28)$$

Dividing the entire equation by the common sign factor $(-1)^{\frac{n+1-m}{2}}$ transforms the addition into a difference equation:

$$(n+1) A_{n+1}(m) - (n-1) A_{n-1}(m) = m A_n(m-1) \quad (1.2.29)$$

Let $N = n+1$, ensuring $N \equiv m \pmod{2}$. The difference equation simplifies to:

$$N A_N(m) - (N-2) A_{N-2}(m) = m A_{N-1}(m-1) \quad (1.2.30)$$

Summing this difference equation from the minimal non-zero index yields the summation operators.

For even powers ($N = 2n$), we obtain:

$$2nA_{2n}(m) = m \sum_{k=1}^n A_{2k-1}(m-1) \quad (1.2.31)$$

For odd powers ($N = 2n + 1$), we obtain:

$$(2n + 1)A_{2n+1}(m) = m \sum_{k=0}^n A_{2k}(m-1) \quad (1.2.32)$$

The base case $m = 1$ corresponds to the standard Maclaurin series for $\arctan(z)$, which confirms $A_{2n+1}(1) = \frac{1}{2n+1}$. By induction, iterating these operators $m - 1$ times accumulates the constant multipliers as $m!$ and generates an $(m - 1)$ -fold nested sum of harmonic terms alternating between even and odd denominators, completing the proof. $\square$

Remark 1.2.2. The Maclaurin expansion for the inverse hyperbolic tangent power, $\operatorname{arctanh}^m(z)$, can be derived as an immediate corollary of 1.2.3. By utilizing the complex identity $\operatorname{arctanh}(z) = -i \arctan(iz)$, we can get the coefficients directly.

Substituting iz into the expansion from (1.2.21) and multiplying by $(-i)^m$ yields:

$$\begin{aligned} \operatorname{arctanh}^m(z) &= (-i)^m \sum_{\substack{N=m \\ N \equiv m \pmod{2}}}^{\infty} (-1)^{\frac{N-m}{2}} A_N(m) (iz)^N \\ &= \sum_{\substack{N=m \\ N \equiv m \pmod{2}}}^{\infty} (-1)^{\frac{N-m}{2}} \left[(-1)^{\frac{N-m}{2}} A_N(m) \right] z^N. \end{aligned}$$

Since $\left((-1)^{\frac{N-m}{2}}\right)^2 = 1$ for all integers $N \equiv m \pmod{2}$, the alternating sign is completely neutralized. Consequently, the Maclaurin expansion for $\operatorname{arctanh}^m(z)$ is generated by the exact same general term $A_N(m)$, but entirely positive terms:

$$\operatorname{arctanh}^m(z) = \sum_{\substack{N=m \\ N \equiv m \pmod{2}}}^{\infty} A_N(m) z^N. \quad (1.2.33)$$

Chapter 2

Asymptotic expansions of gamma function and its different combinations in complex fields

2.1 Introduction

In this chapter, we mainly concern the asymptotic expansions of the Gamma function and its different combinations in complex fields, which are fundamental special functions in mathematics and have wide applications in various fields such as number theory, combinatorics, and mathematical physics.

The Gamma function, typically denoted as $\Gamma(z)$, is one of the most fundamental special functions in mathematics. For large values of $|z|$, the asymptotic behavior of the Gamma function is classically described by Stirling's formula, which states that as $z \rightarrow \infty$ in the sector $|\arg z| < \pi$:

$$\Gamma(z+1) \sim \sqrt{2\pi z} \left(\frac{z}{e}\right)^z. \quad (2.1.1)$$

While Stirling's formula is ubiquitous, a rich history of mathematical literature has been devoted to finding more accurate approximations and asymptotic expansions. Notable historical improvements include Burnside's approximation [7], which shifts the argument to yield better precision, and Ramanujan's remarkable formula [12]:

$$\Gamma(x+1) \sim \sqrt{\pi} \left(\frac{x}{e}\right)^x \left(8x^3 + 4x^2 + x + \frac{1}{30}\right)^{1/6}, \quad x \rightarrow \infty. \quad (2.1.2)$$

These classical results have inspired numerous generalizations, often referred to as Ramanujan-type or Burnside-type asymptotic expansions [11].

Another highly accurate and elegant approximation for the Gamma function was given by Smith [13], who utilized the hyperbolic tangent function to approximate $\Gamma(x+1/2)$:

$$\Gamma\left(x + \frac{1}{2}\right) \sim \sqrt{2\pi} \left(\frac{x}{e}\right)^x \left(2x \tanh \frac{1}{2x}\right)^{x/2}. \quad (2.1.3)$$

Besides these classical approximations, Windschitl also give some very accurate approximations for the Gamma functions, we also derive the complex field case of several Windschitl-type asymptotic expansions.

Motivated by these developments, the first purpose of this part is to establish several new Ramanujan-type asymptotic expansions for the Gamma function $\Gamma(z + 1/2)$ as $z \rightarrow \infty$ in the complex sector $|\arg z| < \pi$. By utilizing series transformations and properties of Bernoulli numbers, we systematically derive explicit recurrence relations for the coefficients of these expansions based on Smith's formula. Furthermore, by manipulating the classical asymptotic series of the Gamma function, we derive expansions related to Burnside's approximation formula and Windschitl's approximation formulas.

Then comes the second part of this chapter, asymptotic expansions of Wallis-type and Gurland ratios of gamma functions in complex fields.

The Wallis quotient function $\frac{\Gamma(z+u)}{\Gamma(z+v)}$ is very useful and frequently occurring in many mathematical problems of analysis. Because of this, many mathematicians are interested in the asymptotic expansion of this quotient or the more generalized form $\left[\frac{\Gamma(z+u)}{\Gamma(z+v)}\right]^{u-v}$ (Wallis power function). In this paper, we may started with some great expansions for the Wallis quotient functions.

There have been two basic forms of the asymptotic expansion of the Wallis quotient function. The first was derived in 1951 by Tricomi and Erdelyi [15], which states that:

$$\frac{\Gamma(z+u)}{\Gamma(z+v)} \sim \sum_{k=0}^{\infty} \frac{(-1)^k \Gamma(v-u+k)}{\Gamma(v-u)k!} B_k^{(v-u+1)}(u) z^{u-v-k}$$

as $z \rightarrow \infty$ with $|\arg(z+u)| < \pi$, where $B_k^{(v-u+1)}(u)$ are the generalized Bernoulli polynomials defined by

$$\left(\frac{t}{e^t - 1}\right)^q e^{xt} = \sum_{i=0}^{\infty} \frac{t^i}{i!} B_k^{(q)}(x), |t| < \pi.$$

An improvement was derived by Fields [9] in 1964, who showed that

$$\frac{\Gamma(z+u)}{\Gamma(z+v)} \sim \sum_{k=0}^{\infty} \frac{(-1)^k \Gamma(v-u+2k)}{\Gamma(v-u)(2k)!} B_{2k}^{(2\rho)}(\rho) (z+\sigma)^{u-v-2k} \quad (2.1.4)$$

as $z \rightarrow \infty$ with $|\arg(z+u)| < \pi$, $\rho = (1+u-v)/2$, $\sigma = (u+v-1)/2$.

And in 1987, using some properties of completely monotonic functions and enveloping series, Frenzen [10] gave a right proof of Luke's guess about Field's result, and established computable error bounds for the asymptotic expansion of the ratio of two gamma functions for the case of real bounded u, v and real z . More precisely, he proved that for $0 < v-u < 1, n \in \mathbb{N}$, and $x+\sigma > 0$,

$$I_n(x) = \frac{\Gamma(x+u)}{\Gamma(x+v)} - \sum_{k=0}^n \frac{b_k}{(x+\sigma)^{v-u+2k}} = \frac{a_{n+1}}{\Gamma(v-u)} \int_0^{\infty} e^{-(x+\sigma)t} t^{2n+2-\rho} \theta_n(t) dt,$$

where

$$a_k = \frac{B_{2k}^{(2\rho)}(\rho)}{(2k)!}, b_k = \frac{\Gamma(v-u+2k)}{\Gamma(v-u)} a_k, \theta_n(t) \in (0, 1)$$

which yields:

$$|I_n(x)| \leq \frac{|a_{n+1}|}{\Gamma(v-u)} \int_0^{\infty} e^{-(x+\sigma)t} t^{2n+2-\rho} \theta_n(t) dt = |b_{n+1}| (x+\sigma)^{u-v-2n-2}.$$

Then, Frenzen's result implies that for any integer $n \geq 0$, the functions $-I_{2n}(x)$ and $I_{2n+1}(x)$ both are completely monotonic on $(-\sigma, \infty)$, and therefore, the double inequality

$$\sum_{k=0}^{2n+1} \frac{b_k}{(x+\sigma)^{v-u+2k}} < \frac{\Gamma(x+u)}{\Gamma(x+v)} < \sum_{k=0}^{2n} \frac{b_k}{(x+\sigma)^{v-u+2k}} \quad (2.1.5)$$

holds for $x > -\sigma$. Therefore, Frenzen's result is excellent.

The second form of the asymptotic expansion of the ratio of two gamma functions was obtained by Burić and Elezović [5], [6], who proved that:

$$\left[\frac{\Gamma(z+u)}{\Gamma(z+v)} \right]^{\frac{1}{u-v}} \sim z \exp\left(\sum_{n=1}^{\infty} \frac{P_n(u, v)}{z^n} \right) \quad (2.1.6)$$

as $z \rightarrow \infty$ with $|\arg(z+u)| < \pi$, where $P_n(u, v)$ are polynomials of order n defined by

$$P_n(u, v) = \frac{(-1)^{n+1}}{n(n+1)} \frac{B_{n+1}(u) - B_{n+1}(v)}{u - v}, \quad (2.1.7)$$

where $B_n(t) = B_n^{(1)}(t)$ stands for the Bernuolli polynomials defined by

$$\frac{xe^{tx}}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n(t)}{n!} x^n$$

as $|x| < 2\pi$.

Recently, a new asymptotic expansion was derived in [17] by Yang, Tian and Ha from Burić and Elezović's expansion [5], [6], who showed that

$$\left[\frac{\Gamma(x+u)}{\Gamma(x+v)} \right]^{\frac{1}{u-v}} \sim (x+\sigma) \exp \left[\sum_{n=1}^{\infty} \frac{B_{2n+1}(\rho)}{\omega n(2n+1)} (x+\sigma)^{-2n} \right]$$

holds as $x \rightarrow \infty$

Inspired by Yang, Tian and Ha's expansions in [17] and Frenzen's great result in [10], we want to find more from Burić and Elezović's 2.1.6, which will improve its computational efficiency. Specifically, we establish the follwing asymptotic expansion:

$$\left[\frac{\Gamma(z+u)}{\Gamma(z+v)} \right]^{\frac{1}{u-v}} \sim (z+\sigma) \exp \left[\sum_{n=1}^{\infty} \frac{B_{2n+1}(\rho)}{\omega n(2n+1)} (z+\sigma)^{-2n} \right]$$

as $z \rightarrow \infty$ with $|\arg(z+u)| < \pi$.

For the Gurland-type ratio of gamma functions, it occupies a prominent place. It is defined by

$$T(x, y) = \frac{\Gamma(x) \Gamma(y)}{\Gamma\left(\frac{x+y}{2}\right)^2}, \quad x, y > 0. \quad (2.1.8)$$

This ratio arises naturally in the study of inequalities between means of Gamma functions, in approximation theory, and in actuarial and statistical applications.

The study of $T(x, y)$ can be traced back at least to the work of Watson, who employed Gauss's hypergeometric formula [1]

$$\frac{\Gamma(c) \Gamma(c-a-b)}{\Gamma(c-a) \Gamma(c-b)} = {}_2F_1(a, b; c; 1), \quad \operatorname{Re}(c-a-b) > 0, \quad (2.1.9)$$

to derive approximation formulas for the Gurland ratio.

In this chapter, we will give an asymptotic expansion of the generalized Gurland-type ratio $\frac{\Gamma(z+p)\Gamma(z+q)}{\Gamma(z+(p+q)/2)^2}$ as $z \rightarrow \infty$ in the complex field.

2.2 Ramanujan-type, Burnside-type and Windschit-type asymptotic expansions of gamma functions

We first give the following lemma, which will be used in the proof of Theorem 2.2.1.

Lemma 2.2.1. If for $z \in \mathbb{C}$ and $|z| < \pi/2$, then

$$\ln \frac{\tanh z}{z} = - \sum_{n=1}^{\infty} \frac{2^{2n} (2^{2n-1} - 1) B_{2n}}{n(2n)!} z^{2n} =: \sum_{n=1}^{\infty} \ddot{a}_n z^{2n}. \quad (2.2.1)$$

Proof. It is well known that

$$\frac{1}{\sinh w} = - \sum_{n=0}^{\infty} \frac{2 (2^{2n-1} - 1) B_{2n}}{(2n)!} w^{2n-1}, \quad |w| < \pi$$

holds (note that the $n = 0$ term corresponds to $1/w$). Thus, for $|2z| < \pi$, which implies $|z| < \pi/2$, we have

$$\begin{aligned} \ln \frac{\tanh z}{z} &= \int_0^{2z} \left(\frac{1}{\sinh w} - \frac{1}{w} \right) dw \\ &= \int_0^{2z} \left(- \sum_{n=1}^{\infty} \frac{2 (2^{2n-1} - 1) B_{2n}}{(2n)!} w^{2n-1} \right) dw \\ &= - \sum_{n=1}^{\infty} \frac{2^{2n} (2^{2n-1} - 1) B_{2n}}{n(2n)!} z^{2n}, \end{aligned}$$

and therefore 2.2.1 holds. $\square$ $\square$

Theorem 2.2.1. As $|z| \rightarrow \infty$ with $z \in \mathbb{C}$ and $|\arg z| < \pi$, the following asymptotic expansion holds:

$$\begin{aligned} \Gamma \left(z + \frac{1}{2} \right) &\sim \sqrt{2\pi} \left(\frac{z}{e} \right)^z \left(2z \tanh \frac{1}{2z} \right)^{z/2} \exp \left(\sum_{n=3}^{\infty} \frac{a_n}{z^{2n-1}} \right) \\ &= \sqrt{2\pi} \left(\frac{z}{e} \right)^z \left(2z \tanh \frac{1}{2z} \right)^{z/2} \\ &\quad \times \exp \left(-\frac{31}{51\,840z^5} + \frac{1397}{2\,419\,200z^7} - \frac{10\,439}{12\,441\,600z^9} + \cdots \right), \end{aligned} \quad (2.2.2)$$

where a_n is defined by the relation derived below.

Proof. From the well-known asymptotic expansion for the Gamma function [11], we have:

$$\ln \Gamma(z + a) = \left(z + a - \frac{1}{2} \right) \ln z - z + \frac{1}{2} \ln 2\pi + \sum_{k=1}^n \frac{(-)^{k+1} B_{k+1}(a)}{k(k+1)} z^{-k} + O(z^{-n-1}), \quad (2.2.3)$$

where $|\arg z| < \pi - \varepsilon$, $\varepsilon > 0$, and the coefficients $\dot{a}_n$ and $\ddot{a}_n$ are defined as:

$$\dot{a}_n = -\frac{(1 - 2^{1-2n})B_{2n}}{2n(2n-1)}, \quad \ddot{a}_n = -\frac{(2^{2n-1} - 1)B_{2n}}{n(2n)!}. \quad (2.2.4)$$

Using Lemma 2.2.1 (with the substitution $t = 1/2z$), we have:

$$\frac{z}{2} \ln \left(2z \tanh \frac{1}{2z} \right) = \frac{1}{2} \sum_{n=1}^{\infty} \frac{\ddot{a}_n}{z^{2n-1}}.$$

Let us define $s_0(z)$ as the exponent in the target expansion:

$$\Gamma\left(z + \frac{1}{2}\right) \sim \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(2z \tanh \frac{1}{2z}\right)^{z/2} \exp(s_0(z)).$$

Then, as $z \rightarrow \infty$, we have:

$$\begin{aligned} s_0(z) &= \left(\ln \Gamma\left(z + \frac{1}{2}\right) - \ln \sqrt{2\pi} - z \ln z + z \right) - \frac{z}{2} \ln \left(2z \tanh \frac{1}{2z}\right) \\ &\sim \sum_{n=1}^{\infty} \frac{\dot{a}_n}{z^{2n-1}} - \frac{1}{2} \sum_{n=1}^{\infty} \frac{\ddot{a}_n}{z^{2n-1}} = \sum_{n=1}^{\infty} \frac{\dot{a}_n - \ddot{a}_n/2}{z^{2n-1}} = \sum_{n=1}^{\infty} \frac{a_n}{z^{2n-1}}. \end{aligned} \quad (2.2.5)$$

Through simple calculation, we find that $a_1 = a_2 = 0$, and

$$a_3 = -\frac{31}{51\,840}, \quad a_4 = \frac{1397}{2\,419\,200}, \quad a_5 = -\frac{10\,439}{12\,441\,600}.$$

Thus, the conclusion is proved. $\square$

Next, we present the second Ramanujan-type asymptotic expansion related to Smith's approximation formula for the Gamma function.

Since the Bernoulli numbers $B_{2n+1} = 0$ for $n \in \mathbb{N}$, the asymptotic series $s_0(z)$ given previously can be rewritten in the following form:

$$s_0(z) = \sum_{n=1}^{\infty} \frac{(2^n - (n+1)(n-1)!(1-2^{-n})B_{n+1})}{(n+1)(n+1)!} \frac{1}{z^n} =: \sum_{n=1}^{\infty} \frac{a_n^*}{z^n}, \quad (2.2.6)$$

where

$$a_n^* = \frac{(2^n - (n+1)(n-1)!(1-2^{-n})B_{n+1})}{(n+1)(n+1)!} B_{n+1}. \quad (2.2.7)$$

Obviously, for $1 \leq n \leq 4$, we have $a_n^* = 0$.

Thus, by means of series transformation, we can establish the following second Ramanujan-type asymptotic expansion for the Gamma function $\Gamma(z + 1/2)$.

Before stating the main result, we first present a lemma that will be used in the proof of Theorem 2.2.2.

Lemma 2.2.2. Let $A(z)$ be a function with an asymptotic expansion as $z \rightarrow \infty$ and $|\arg z| < \pi$:

$$A(z) \sim \sum_{n=1}^{\infty} a_n z^{-n}. \quad (2.2.8)$$

Then the composition $B(z) = \exp(A(z))$ has an asymptotic expansion of the following form:

$$B(z) \sim \sum_{n=0}^{\infty} b_n z^{-n}, \quad (2.2.9)$$

where $b_0 = 1$ and

$$b_n = \frac{1}{n} \sum_{k=1}^n k a_k b_{n-k}, \quad n \geq 1. \quad (2.2.10)$$

Proof. The existence of the asymptotic expansion of the function $B(z)$ follows from the standard properties of the composition of asymptotic expansions. Suppose for the moment that $A(z)$ is analytic and that its derivative $A'(z)$ has an asymptotic expansion obtained by term-by-term differentiation.

Differentiating the equation $B(z) = \exp(A(z))$ yields

$$B'(z) = A'(z)e^{A(z)} = A'(z)B(z),$$

so $B'(z)$ also has an asymptotic expansion. Utilizing the Cauchy product of the series, we can write

$$\begin{aligned} B'(z) &\sim \left(\sum_{n=1}^{\infty} a_n(-n)z^{-n-1} \right) \left(\sum_{n=0}^{\infty} b_n z^{-n} \right) \\ &\sim \sum_{n=0}^{\infty} \left(\sum_{k=0}^n (-k)a_k b_{n-k} \right) z^{-n-1}. \end{aligned}$$

But we also have the formal differentiation of $B(z)$ itself:

$$B'(z) \sim \sum_{n=0}^{\infty} b_n(-n)z^{-n-1}.$$

Equating the coefficients of z^{-n-1} from both expressions implies that for $n \geq 1$,

$$-nb_n = \sum_{k=0}^n (-k)a_k b_{n-k} \implies b_n = \frac{1}{n} \sum_{k=1}^n k a_k b_{n-k}.$$

Let us show that the assumption about the existence of the term-by-term differentiated asymptotic expansion of the function $A(z)$ is not necessary. Choose N arbitrarily large and define the truncated series

$$\begin{aligned} \tilde{A}(z) &= \sum_{n=1}^N a_n z^{-n}, \\ \tilde{B}(z) &= \exp(\tilde{A}(z)). \end{aligned}$$

Then the first N coefficients of the asymptotic expansion of the function $\tilde{B}(z)$ coincide with the coefficients of the function $B(z)$. Furthermore, $\tilde{A}(z)$ is an analytic function (specifically, a polynomial in $1/z$), and its derivative $\tilde{A}'(z)$ naturally possesses an asymptotic expansion. From the equation $\tilde{B}'(z) = \tilde{A}'(z)\tilde{B}(z)$, one obtains the identical relationship between the coefficients for all indices $n \leq N$. Since N can be chosen to be arbitrarily large, this finishes the proof. $\square$

Theorem 2.2.2. *For $r \neq 0$, as $|z| \rightarrow \infty$ with $|\arg z| < \pi$, the following asymptotic expansion holds:*

$$\Gamma\left(z + \frac{1}{2}\right) \sim \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(2z \tanh \frac{1}{2z}\right)^{z/2} \left(1 + \sum_{n=1}^{\infty} \frac{b_n}{z^n}\right)^r, \quad (2.2.11)$$

where $b_0 = 1$, $b_1 = b_2 = b_3 = b_4 = 0$, and for $n \geq 5$,

$$b_n = \frac{1}{rn} \sum_{k=5}^n \frac{(2^k k - (k+1)!)(1 - 2^{-k})}{(k+1)(k+1)!} B_{k+1} b_{n-k}. \quad (2.2.12)$$

Proof. According to Lemma 2.2.2, if as $|z| \rightarrow \infty$ we have

$$\exp\left(\sum_{n=1}^{\infty} a_n z^{-n}\right) \sim \sum_{n=0}^{\infty} b_n z^{-n},$$

then the coefficients satisfy $b_0 = 1$ and $b_n = \frac{1}{n} \sum_{k=1}^n k a_k b_{n-k}$ for $n \geq 1$. Thus, for $r \neq 0$, if as $|z| \rightarrow \infty$ we have

$$\exp \left(\sum_{n=1}^{\infty} a_n z^{-n} \right) \sim \left(\sum_{n=0}^{\infty} b_n z^{-n} \right)^r,$$

which is equivalent to

$$\exp \left(\sum_{n=1}^{\infty} \frac{a_n}{r} z^{-n} \right) \sim \sum_{n=0}^{\infty} b_n z^{-n},$$

then $b_0 = 1$, and for $n \geq 1$,

$$b_n = \frac{1}{rn} \sum_{k=1}^n k a_k b_{n-k}. \quad (2.2.13)$$

Substituting a_k^* (as defined in the previous derivation for $s_0(z)$) into the above recurrence relation, nd noting that $a_k^* = 0$ for $1 \leq k \leq 4$, we obtain the recurrence formula for b_n as stated. The conclusion follows. $\square$

Setting $r = 1$ in Theorem 2.2.2, we note that

$$\begin{aligned} b_5 &= -\frac{31}{51840}, & b_6 &= 0, & b_7 &= \frac{1397}{2419200}, & b_8 &= 0, & b_9 &= -\frac{10439}{12441600}, \\ b_{10} &= \frac{961}{5374771200}, & b_{11} &= \frac{4626754267}{2414168064000}, & b_{12} &= -\frac{43307}{125411328000}, \end{aligned}$$

and the following corollary is immediate.

Corollary 2.2.3. *As $z \rightarrow \infty$ and $|\arg z| < \pi$, the asymptotic expansion*

$$\begin{aligned} \Gamma \left(z + \frac{1}{2} \right) &\sim \sqrt{2\pi} \left(\frac{z}{e} \right)^z \left(2z \tanh \frac{1}{2z} \right)^{z/2} \left(1 + \sum_{n=5}^{\infty} \frac{b_n}{z^n} \right) \\ &= \sqrt{2\pi} \left(\frac{z}{e} \right)^z \left(2z \tanh \frac{1}{2z} \right)^{z/2} \left(1 - \frac{31}{51840 z^5} + \frac{1397}{2419200 z^7} - \frac{10439}{12441600 z^9} \right. \\ &\quad \left. + \frac{961}{5374771200 z^{10}} + \frac{4626754267}{2414168064000 z^{11}} - \frac{43307}{125411328000 z^{12}} + \dots \right) \end{aligned}$$

holds, where $b_0 = 1$, $b_n = 0$ for $1 \leq n \leq 4$, and for $n \geq 5$,

$$b_n = \frac{1}{n} \sum_{k=5}^n \frac{(k2^k - (k+1)!) (1 - 2^{-k})}{(k+1)(k+1)!} B_{k+1} b_{n-k}.$$

Setting $r = 1/6$ in Theorem 2.2.2, we note that

$$\begin{aligned} b_5 &= -\frac{31}{8640}, & b_6 &= 0, & b_7 &= \frac{1397}{403200}, & b_8 &= 0, & b_9 &= -\frac{10439}{2073600}, \\ b_{10} &= \frac{961}{149299200}, & b_{11} &= \frac{4626754267}{402361344000}, & b_{12} &= -\frac{43307}{3483648000}, \end{aligned}$$

and the following corollary holds.

Corollary 2.2.4. *As $z \rightarrow \infty$ and $|\arg z| < \pi$, the asymptotic expansion*

$$\begin{aligned} \Gamma\left(z + \frac{1}{2}\right) &\sim \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(2z \tanh \frac{1}{2z}\right)^{z/2} \left(1 + \sum_{n=5}^{\infty} \frac{b_n}{z^n}\right)^{1/6} \\ &= \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(2z \tanh \frac{1}{2z}\right)^{z/2} \left(1 - \frac{31}{8640z^5} + \frac{1397}{403200z^7} - \frac{10439}{2073600z^9} \right. \\ &\quad \left. + \frac{961}{149299200z^{10}} + \frac{4626754267}{402361344000z^{11}} - \frac{43307}{3483648000z^{12}} + \cdots\right)^{1/6} \end{aligned}$$

holds, where $b_0 = 1$, $b_n = 0$ for $1 \leq n \leq 4$, and for $n \geq 5$,

$$b_n = \frac{6}{n} \sum_{k=5}^n \frac{(k2^k - (k+1)!) (1 - 2^{-k})}{(k+1)(k+1)!} B_{k+1} b_{n-k}.$$

Since $B_{2n+1} = 0$ for $n \in \mathbb{N}$, the asymptotic series in (2.2.3) can be written as

$$-\sum_{n=1}^{\infty} \frac{(1 - 2^{1-2n}) B_{2n}}{2n(2n-1)} \cdot \frac{1}{z^{2n-1}} = -\sum_{m=1}^{\infty} \frac{(1 - 2^{-m}) B_{m+1}}{m(m+1)} \cdot \frac{1}{z^m} = -\sum_{n=1}^{\infty} \frac{\delta_n}{z^n},$$

where

$$\delta_n = \frac{(1 - 2^{-n}) B_{n+1}}{n(n+1)}.$$

Thus, by applying a proof method analogous to that of Theorem 2.2.2, we obtain the following third Ramanujan-type asymptotic expansion for the gamma function $\Gamma(z + 1/2)$.

Theorem 2.2.5. *For $r \neq 0$, as $z \rightarrow \infty$ and $|\arg z| < \pi$, the asymptotic expansion*

$$\Gamma\left(z + \frac{1}{2}\right) \sim \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(1 + \sum_{n=1}^{\infty} \frac{c_n}{z^n}\right)^r$$

holds, where $c_0 = 1$, and for $n \geq 1$,

$$c_n = -\frac{1}{rn} \sum_{k=1}^n \frac{(1 - 2^{-k}) B_{k+1}}{k+1} c_{n-k}.$$

In particular, taking $r = 1, 1/6, 1/8, 1/12$ in Theorem 2.2.5 respectively, the following corollaries hold.

Corollary 2.2.6. *As $z \rightarrow \infty$ and $|\arg z| < \pi$, the asymptotic expansion*

$$\begin{aligned} \Gamma\left(z + \frac{1}{2}\right) &\sim \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(1 + \sum_{n=1}^{\infty} \frac{c_n}{z^n}\right) \\ &= \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(1 - \frac{1}{24z} + \frac{1}{1152z^2} + \frac{1003}{414720z^3} - \frac{4027}{39813120z^4} + \cdots\right) \end{aligned}$$

holds, where $c_0 = 1$, and for $n \geq 1$,

$$c_n = -\frac{1}{n} \sum_{k=1}^n \frac{(1 - 2^{-k}) B_{k+1}}{k+1} c_{n-k}.$$

Corollary 2.2.7. *As $z \rightarrow \infty$ and $|\arg z| < \pi$, the asymptotic expansion*

$$\begin{aligned}\Gamma\left(z + \frac{1}{2}\right) &\sim \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(1 + \sum_{n=1}^{\infty} \frac{c_n}{z^n}\right)^{1/6} \\ &= \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(1 - \frac{1}{4z} + \frac{1}{32z^2} + \frac{23}{1920z^3} - \frac{107}{30720z^4} + \cdots\right)^{1/6}\end{aligned}$$

holds, where $c_0 = 1$, and for $n \geq 1$,

$$c_n = -\frac{6}{n} \sum_{k=1}^n \frac{(1 - 2^{-k}) B_{k+1}}{k+1} c_{n-k}.$$

Corollary 2.2.8. *As $z \rightarrow \infty$ and $|\arg z| < \pi$, the asymptotic expansion*

$$\begin{aligned}\Gamma\left(z + \frac{1}{2}\right) &\sim \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(1 + \sum_{n=1}^{\infty} \frac{c_n}{z^n}\right)^{1/8} \\ &= \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(1 - \frac{1}{3z} + \frac{1}{18z^2} + \frac{43}{3240z^3} - \frac{29}{4860z^4} + \cdots\right)^{1/8}\end{aligned}$$

holds, where $c_0 = 1$, and for $n \geq 1$,

$$c_n = -\frac{8}{n} \sum_{k=1}^n \frac{(1 - 2^{-k}) B_{k+1}}{k+1} c_{n-k}.$$

Corollary 2.2.9. *As $z \rightarrow \infty$ and $|\arg z| < \pi$, the asymptotic expansion*

$$\begin{aligned}\Gamma\left(z + \frac{1}{2}\right) &\sim \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(1 + \sum_{n=1}^{\infty} \frac{c_n}{z^n}\right)^{1/12} \\ &= \sqrt{2\pi} \left(\frac{z}{e}\right)^z \left(1 - \frac{1}{2z} + \frac{1}{8z^2} + \frac{1}{120z^3} - \frac{23}{1920z^4} + \cdots\right)^{1/12}\end{aligned}$$

holds, where $c_0 = 1$, and for $n \geq 1$,

$$c_n = -\frac{12}{n} \sum_{k=1}^n \frac{(1 - 2^{-k}) B_{k+1}}{k+1} c_{n-k}.$$

Having established several Ramanujan-type results, we next present an asymptotic expansion related to Burnside's classical approximation. By manipulating the standard asymptotic series of the Gamma function, we derive the following Burnside-type expansion.

Theorem 2.2.10. *As $z \rightarrow \infty$ with $z \in \mathbb{C}$ and $|\arg z| < \pi$, the following asymptotic expansion holds:*

$$\Gamma\left(z + \frac{1}{2}\right) \sim \sqrt{2\pi} \left(\frac{z}{e}\right)^z \exp\left(-\frac{1}{24z} \frac{1}{\sum_{n=0}^{\infty} c_n z^{-2n}}\right), \quad (2.2.14)$$

where $c_0 = 1$, and for $n \geq 1$,

$$c_n = -24 \sum_{k=1}^n \frac{(1 - 2^{-2k-1}) B_{2k+2}}{2(k+1)(2k+1)} c_{n-k}, \quad (2.2.15)$$

where B_{2k+2} are the Bernoulli numbers.

Proof. As $z \rightarrow \infty$, the asymptotic expansion is equivalent to

$$-24z \left(\ln \Gamma \left(z + \frac{1}{2} \right) - \ln \sqrt{2\pi} - z \ln z + z \right) \sum_{n=0}^{\infty} c_n z^{-2n} \sim 1.$$

Then, as $z \rightarrow \infty$, from the known asymptotic expansion of the gamma function [2.2.3](#), we have

$$24z \sum_{n=1}^{\infty} \frac{(1 - 2^{1-2n}) B_{2n}}{2n(2n-1)} \frac{1}{z^{2n-1}} \sum_{n=0}^{\infty} c_n z^{-2n} \sim 1,$$

which can be rewritten by changing the summation index as

$$24 \sum_{n=0}^{\infty} \left(\sum_{k=0}^n \frac{(1 - 2^{-2k-1}) B_{2k+2}}{2(k+1)(2k+1)} c_{n-k} \right) \frac{1}{z^{2n}} \sim 1.$$

Equating the coefficients of z^{-2n} on both sides, we obtain $c_0 = 1$ for $n = 0$, and for $n \geq 1$,

$$\sum_{k=0}^n \frac{(1 - 2^{-2k-1}) B_{2k+2}}{2(k+1)(2k+1)} c_{n-k} = 0.$$

Rearranging this yields the recurrence relation for c_n . Thus, the conclusion is proved. $\square$

Next we will give two Windschit-type asymptotic expansions for the gamma function $\Gamma(z+1)$.

Before stating the main result, we first present a lemma that will be used in the proof of the following theorem.

Lemma 2.2.3. For $|z| < \pi$, we have

$$\ln \frac{\sinh z}{z} = \sum_{n=1}^{\infty} \frac{2^{2n} B_{2n}}{2n(2n)!} z^{2n}. \quad (2.2.16)$$

Proof. It is well known that

$$\frac{1}{\tanh w} = \sum_{n=0}^{\infty} \frac{2^{2n} B_{2n}}{(2n)!} w^{2n-1}, \quad 0 < |w| < \pi,$$

where the $n = 0$ term corresponds to $1/w$. Thus, for $|z/2| < \pi$, which implies $|z| < \pi$, we have

$$\begin{aligned} \ln \frac{\sinh z}{z} &= \int_0^z \left(\frac{1}{\tanh w} - \frac{1}{w} \right) dw \\ &= \int_0^z \left(\sum_{n=1}^{\infty} \frac{2^{2n} B_{2n}}{(2n)!} w^{2n-1} \right) dw \\ &= \sum_{n=1}^{\infty} \frac{2^{2n} B_{2n}}{2n(2n)!} z^{2n}, \end{aligned}$$

and therefore [\(2.2.16\)](#) holds. $\square$

Theorem 2.2.11. As $|z| \rightarrow \infty$ with $|\arg z| < \pi$, the asymptotic expansion

$$\begin{aligned} \Gamma(z+1) &\sim \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \left(z \sinh \frac{1}{z}\right)^{z/2} \exp\left(\sum_{n=3}^{\infty} \frac{a_n}{z^{2n-1}}\right) \\ &= \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \left(z \sinh \frac{1}{z}\right)^{z/2} \\ &\quad \times \exp\left[\frac{1}{1620z^5} - \frac{11}{18900z^7} + \frac{143}{170100z^9} - \frac{2260261}{1178793000z^{11}} + \cdots\right] \end{aligned} \quad (2.2.17)$$

holds, where

$$a_n = \frac{2n(2n-2)! - 2^{2n-1}}{2n(2n)!} B_{2n}, \quad (2.2.18)$$

and B_{2n} are the Bernoulli numbers.

Proof. From the asymptotic expansion [11] and Lemma 2.2.3, as $z \rightarrow \infty$, we have

$$\ln \Gamma(z+1) - \ln \sqrt{2\pi} - \left(z + \frac{1}{2}\right) \ln z + z \sim \sum_{n=1}^{\infty} \frac{a'_n}{z^{2n-1}} \quad (2.2.19)$$

and

$$\frac{z}{2} \ln \left(z \sinh \frac{1}{z}\right) = \frac{1}{2} \sum_{n=1}^{\infty} \frac{a''_n}{z^{2n-1}} \quad (2.2.20)$$

hold, where

$$a'_n = \frac{B_{2n}}{2n(2n-1)}, \quad a''_n = \frac{2^{2n} B_{2n}}{2n(2n)!}.$$

As $z \rightarrow \infty$, let

$$\Gamma(z+1) \sim \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \left(z \sinh \frac{1}{z}\right)^{z/2} \exp(w_0(z)).$$

Thus, as $z \rightarrow \infty$, we have

$$\begin{aligned} w_0(z) &= \left(\ln \Gamma(z+1) - \ln \sqrt{2\pi} - \left(z + \frac{1}{2}\right) \ln z + z\right) - \frac{z}{2} \ln \left(z \sinh \frac{1}{z}\right) \\ &\sim \sum_{n=1}^{\infty} \frac{a'_n}{z^{2n-1}} - \frac{1}{2} \sum_{n=1}^{\infty} \frac{a''_n}{z^{2n-1}} = \sum_{n=1}^{\infty} \frac{a'_n - a''_n/2}{z^{2n-1}} = \sum_{n=1}^{\infty} \frac{a_n}{z^{2n-1}}. \end{aligned}$$

Through some simple calculations, we obtain $a_1 = a_2 = 0$, and

$$a_3 = \frac{1}{1620}, \quad a_4 = -\frac{11}{18900}, \quad a_5 = \frac{143}{170100}, \quad a_6 = -\frac{2260261}{1178793000}.$$

Thus, the conclusion holds. $\square$

Remark 2.2.1. Since $B_{2n+1} = 0$ for $n \in \mathbb{N}$, the asymptotic series $w_0(z)$ can also be expressed in the following form:

$$\begin{aligned} w_0(z) &= \sum_{n=3}^{\infty} \frac{2n(2n-2)! - 2^{2n-1}}{2n(2n)!} \frac{B_{2n}}{z^{2n-1}} \\ &= \sum_{n=1}^{\infty} \frac{(n+1)(n-1)! - 2^n}{(n+1)(n+1)!} \frac{B_{n+1}}{z^n} := \sum_{n=1}^{\infty} \frac{a_n^*}{z^n}, \end{aligned}$$

where

$$a_n^* = \frac{(n+1)(n-1)! - 2^n}{(n+1)(n+1)!} B_{n+1}. \quad (2.2.21)$$

Below we give the second Windschitl-type asymptotic expansion of the Gamma function.

Theorem 2.2.12. *As $z \rightarrow \infty$, the asymptotic expansion*

$$\Gamma(z+1) \sim \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \left(z \sinh \frac{1}{z}\right)^{z/2} \left(1 + \sum_{n=1}^{\infty} \frac{b_n}{z^n}\right) \quad (2.2.22)$$

holds, where $b_0 = 1$, $b_1 = b_2 = b_3 = b_4 = 0$, and

$$b_n = \frac{1}{n} \sum_{k=1}^n \left(\frac{1}{k+1} - \frac{2^k}{(k+1)^2(k-1)!} \right) B_{k+1} b_{n-k} \quad (n \geq 5), \quad (2.2.23)$$

with $|\arg z| \leq \pi - \varepsilon$, $\varepsilon > 0$.

Proof. By Lemma 2.2.2, as $z \rightarrow \infty$ with $|\arg z| \leq \pi - \varepsilon$, $\varepsilon > 0$,

$$\exp \left(\sum_{n=1}^{\infty} a_n z^{-n} \right) \sim \sum_{n=0}^{\infty} b_n z^{-n},$$

where $b_0 = 1$ and

$$b_n = \frac{1}{n} \sum_{k=1}^n k a_k b_{n-k}, \quad n \geq 1. \quad (2.2.24)$$

Combining with (2.2.21) and the above formula, formula (2.2.23) follows by straightforward computation. Furthermore, we have $b_n = 0$ for $1 \leq n \leq 4$, $b_6 = b_8 = 0$, and

$$b_5 = \frac{1}{1620}, \quad b_7 = -\frac{11}{18900}, \quad b_9 = \frac{143}{170100}, \quad b_{10} = \frac{1}{5248800},$$

from which the conclusion follows. $\square$

Now we will state a theorem which is an improvement of [8, Theorem 2], before giving its proof, we will first present the proof of the complex field case of [8, Theorem 2].

We will first give a lemma that will be used in the proof of the following two theorems.

Lemma 2.2.4. Let $c_0 \neq 0$ and let $C(z)$ be a function with an asymptotic expansion as $z \rightarrow \infty$ and $|\arg z| < \pi$:

$$C(z) \sim \sum_{n=0}^{\infty} c_n z^{-n}. \quad (2.2.25)$$

Then the composition $A(z) = \ln(C(z))$ has an asymptotic expansion of the following form:

$$A(z) \sim \sum_{n=1}^{\infty} a_n z^{-n}, \quad (2.2.26)$$

where $a_0 = \ln c_0$ and

$$a_n = \frac{c_n}{c_0} - \frac{1}{nc_0} \sum_{k=1}^{n-1} k a_k c_{n-k}, \quad n \geq 1. \quad (2.2.27)$$

Proof. Since $A(z) = \ln(C(z))$, we have $C(z) = \exp(A(z))$, so $A(z)$ and $C(z)$ are precisely in the setting of Lemma 2.2.2 with the roles of $A(z)$ and $C(z)$ as there. Applying (2.2.9) with $b_n = c_n$ and isolating

the $k = n$ term gives

$$c_n = a_n c_0 + \frac{1}{n} \sum_{k=1}^{n-1} k a_k c_{n-k}, \quad n \geq 1, \quad (2.2.28)$$

and solving for a_n yields (2.2.27) immediately. $\square$

Now, we are ready to state the complex field case for the [8, Theorem 2].

Theorem 2.2.13. *For $z \rightarrow \infty$ with $|\arg z| < \pi$, the gamma function has the following asymptotic expansion:*

$$\Gamma(z+1) \sim \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \left(z \sinh \frac{1}{z}\right)^{z/2 + \sum_{j=0}^{\infty} r_j z^{-j}}, \quad (2.2.29)$$

with the coefficients r_j given by

$$r_j = 6 \left(p_{j+2} - \frac{q_{j+3}}{2} - \sum_{k=0}^{j-1} r_k q_{j-k+2} \right), \quad j \in \mathbb{N}_0, \quad (2.2.30)$$

where

$$p_j = \frac{B_{j+1}}{j(j+1)} \quad \text{and} \quad q_j = c_j - \frac{1}{j} \sum_{k=1}^{j-1} k q_k c_{j-k}, \quad j \in \mathbb{N}, \quad (2.2.31)$$

with

$$c_{2j} = \frac{1}{(2j+1)!} \quad \text{and} \quad c_{2j+1} = 0, \quad j \in \mathbb{N}_0. \quad (2.2.32)$$

Here B_n are the Bernoulli numbers, and an empty sum is understood to be nil.

Proof. Rewrite (2.2.29) as

$$\frac{\ln\left(\frac{\Gamma(z+1)}{\sqrt{2\pi z} (z/e)^z}\right)}{\ln\left(z \sinh \frac{1}{z}\right)} \sim \frac{z}{2} + \sum_{j=0}^{\infty} \frac{r_j}{z^j}, \quad z \rightarrow \infty. \quad (2.2.33)$$

From the well-known asymptotic expansion [11] it follows that

$$\ln\left(\frac{\Gamma(z+1)}{\sqrt{2\pi z} (z/e)^z}\right) \sim \sum_{j=1}^{\infty} \frac{p_j}{z^j}, \quad (2.2.34)$$

with $p_j = B_{j+1}/(j(j+1))$. The Taylor series $\sinh z = \sum_{j=0}^{\infty} \frac{z^{2j+1}}{(2j+1)!}$ gives $z \sinh(1/z) = 1 + \sum_{j=1}^{\infty} c_j z^{-j}$, and applying Lemma 2.2.4 with $c_0 = 1$ yields

$$\ln\left(z \sinh \frac{1}{z}\right) \sim \sum_{j=1}^{\infty} \frac{q_j}{z^j}, \quad (2.2.35)$$

where $q_j = c_j - \frac{1}{j} \sum_{k=1}^{j-1} k q_k c_{j-k}$. Dividing (2.2.34) by (2.2.35) and noting that $q_1 = c_1 = 0$, $p_1 = \frac{1}{12}$ and $q_2 = c_2 = 1/6$, equation (2.2.33) becomes

$$\sum_{j=1}^{\infty} \frac{p_j}{z^j} \sim \left(\frac{z}{2} + \sum_{j=0}^{\infty} \frac{r_j}{z^j} \right) \sum_{k=2}^{\infty} \frac{q_k}{z^k}. \quad (2.2.36)$$

Expanding the right-hand side and matching coefficients of z^{-j} on both sides yields, for $j \geq 2$,

$$p_j - \frac{q_{j+1}}{2} = \sum_{k=0}^{j-2} r_k q_{j-k}, \quad (2.2.37)$$

and re-indexing with $j \mapsto j+2$ gives exactly (2.2.30). $\square$

Finally, we are ready to state the main theorem which is an improvement of [8, Theorem 2].

Theorem 2.2.14. *As $z \rightarrow \infty$ with $|\arg z| \leq \pi - \varepsilon$, $\varepsilon > 0$, the asymptotic expansion*

$$\Gamma(z+1) \sim \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \left(z \sinh \frac{1}{z}\right)^{\frac{z}{2}(1+\sum_{n=2}^{\infty} c_n z^{-2n})} \quad (2.2.38)$$

holds, or equivalently,

$$\Gamma(z+1) \sim \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \left(z \sinh \frac{1}{z}\right)^{\frac{z}{2}\left(1+\frac{1}{135z^4}-\frac{191}{28350z^6}+\frac{25127}{2551500z^8}-\frac{19084273}{841995000z^{10}}+\cdots\right)},$$

where $c_0 = 1$, $c_1 = 0$, and for $n \geq 2$,

$$c_n = \frac{6B_{2n+2}}{(n+1)(2n+1)} - 6 \sum_{k=1}^n \frac{2^{2k+2} B_{2k+2}}{2(k+1)(2k+2)!} c_{n-k}. \quad (2.2.39)$$

Proof. Taking the logarithm of the asymptotic expansion (2.2.29) in the previous theorem, which is valid for $|\arg z| \leq \pi - \varepsilon$, $\varepsilon > 0$, we can rewrite it as

$$\ln \Gamma(z+1) - \ln \sqrt{2\pi z} - z \ln z + z \sim \left(\frac{z}{2} + \sum_{j=0}^{\infty} \frac{r_j}{z^j}\right) \ln \left(z \sinh \frac{1}{z}\right).$$

In view of [11] and 2.2.3, this is equivalent to

$$\sum_{n=1}^{\infty} \frac{B_{2n}}{2n(2n-1)} \frac{1}{z^{2n-1}} \sim \left(\frac{z}{2} + \sum_{j=0}^{\infty} \frac{r_j}{z^j}\right) \left(\sum_{n=1}^{\infty} \frac{2^{2n} B_{2n}}{2n(2n)!} \frac{1}{z^{2n}}\right).$$

Since the left-hand side is an odd function and the second factor on the right-hand side is an even function, the asymptotic series $\frac{z}{2} + \sum_{j=0}^{\infty} r_j z^{-j}$ must be an odd function, and hence $r_{2n} = 0$ for $n \in \mathbb{N} \cup \{0\}$. Consequently, the above can be transformed into:

$$\sum_{n=0}^{\infty} \frac{B_{2n+2}}{(n+1)(2n+1)} \frac{1}{z^{2n}} \sim \left(\sum_{n=0}^{\infty} \frac{c_n}{z^{2n}}\right) \left(\sum_{n=1}^{\infty} \frac{2^{2n+2} B_{2n+2}}{2(n+1)(2n+2)!} \frac{1}{z^{2n}}\right).$$

It follows that

$$\frac{B_{2n+2}}{(n+1)(2n+1)} = \sum_{k=0}^n \frac{2^{2k+2} B_{2k+2}}{2(k+1)(2k+2)!} c_{n-k},$$

hence $c_0 = 1$, and for $n \geq 1$,

$$c_n = \frac{6B_{2n+2}}{(n+1)(2n+1)} - 6 \sum_{k=1}^n \frac{2^{2k+2} B_{2k+2}}{2(k+1)(2k+2)!} c_{n-k}.$$

By straightforward computation,

$$c_1 = 0, \quad c_2 = \frac{1}{135}, \quad c_3 = -\frac{191}{28350}, \quad c_4 = \frac{25127}{2551500}, \quad c_5 = -\frac{19084273}{841995000},$$

from which the conclusion follows. $\square$

Remark 2.2.2. If the series (2.2.17), (2.2.22), (2.2.13) and (2.2.38) are truncated at $n = 3, 5, 3, 2$ respectively, then as $z \rightarrow \infty$ with $|\arg z| < \pi$, the following four Windschitl-type approximations hold:

$$\Gamma(z+1) \sim \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \left(z \sinh \frac{1}{z}\right)^{z/2} \exp\left(\frac{1}{1620z^5}\right) := W_{01}(z), \quad (2.2.40)$$

$$\Gamma(z+1) \sim \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \left(z \sinh \frac{1}{z}\right)^{z/2} \left(1 + \frac{1}{1620z^5}\right) := W_{01}^*(z), \quad (2.2.41)$$

$$\Gamma(z+1) \sim \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \left(z \sinh \frac{1}{z} + \frac{1}{810z^6}\right)^{z/2} =: W_1(z), \quad (2.2.42)$$

$$\Gamma(z+1) \sim \sqrt{2\pi z} \left(\frac{z}{e}\right)^z \left(z \sinh \frac{1}{z}\right)^{\frac{z}{2}\left(1 + \frac{1}{135z^4}\right)} =: W_{c1}(z). \quad (2.2.43)$$

Now, we will give some numerical comparisons of the above four approximations. As $z \rightarrow \infty$ with $|\arg z| < \pi$, the following limits hold:

$$\begin{aligned} \lim_{z \rightarrow \infty} z^7 (\ln \Gamma(z+1) - \ln W_1(z)) &= -\frac{163}{340200}, \\ \lim_{z \rightarrow \infty} z^7 (\ln \Gamma(z+1) - \ln W_{c1}(z)) &= -\frac{191}{340200}, \\ \lim_{z \rightarrow \infty} z^7 (\ln \Gamma(z+1) - \ln W_{01}(z)) &= \lim_{z \rightarrow \infty} z^7 (\ln \Gamma(z+1) - \ln W_{01}^*(z)) = -\frac{198}{340200}. \end{aligned}$$

In particular, since $163 < 191 < 198$, the approximation $W_1(z)$ is the most accurate among the four at order z^{-7} , followed by $W_{c1}(z)$, while $W_{01}(z)$ and $W_{01}^*(z)$ share the same accuracy at this order.

2.3 Asymptotic expansions for the different kinds of ratios of Gamma functions

Now, we will give some Wallis-type asymptotic expansion for the ratio of Gamma functions. First, we give a asymptotic expansion for the ratio $\frac{\Gamma(z+1)}{\Gamma(z+\frac{1}{2})}$.

Theorem 2.3.1. *As $z \rightarrow \infty$ with $|\arg z| < \pi$, the asymptotic expansion*

$$\begin{aligned} \frac{\Gamma(z+1)}{\Gamma(z+1/2)} &\sim \sqrt{z} \left(\cosh \frac{1}{2z}\right)^z \exp\left(\sum_{n=3}^{\infty} \frac{u_n}{z^{2n-1}}\right) \\ &= \sqrt{z} \left(\cosh \frac{1}{2z}\right)^z \exp\left(\frac{7}{5760z^5} - \frac{187}{161280z^7} + \frac{48763}{29030400z^9} - \frac{29383393}{7664025600z^{11}} + \cdots\right) \end{aligned} \quad (2.3.1)$$

holds, where the coefficients are given by

$$u_n = \frac{(2n)! - (2n-1)2^{2n-1}}{n(2n-1)(2n)!} (1 - 2^{-2n}) B_{2n}, \quad (2.3.2)$$

with B_{2n} denoting the Bernoulli numbers.

Proof. The result follows immediately from the asymptotic expansion (2.2.17). $\square$

We now present a Ramanujan-type asymptotic expansion for the Wallis-type ratio of two gamma functions.

Theorem 2.3.2. *Let $r \neq 0$. As $z \rightarrow \infty$ with $|\arg z| < \pi$, the asymptotic expansion*

$$\frac{\Gamma(z+1)}{\Gamma(z+1/2)} \sim \sqrt{z} \left(\cosh \frac{1}{2z} \right)^z \left(1 + \sum_{n=1}^{\infty} \frac{v_n}{z^n} \right)^r \quad (2.3.3)$$

holds, where $v_0 = 1$, $v_n = 0$ for $1 \leq n \leq 4$, and for $n \geq 5$,

$$v_n = \frac{1}{rn} \sum_{k=5}^n \frac{((k+1)! - k2^k)(2 - 2^{-k}) B_{k+1}}{(k+1)(k+1)!} v_{n-k}. \quad (2.3.4)$$

Proof. Since $B_{2n+1} = 0$ for $n \in \mathbb{N}$, the asymptotic series $\sum_{n=3}^{\infty} u_n z^{1-2n}$ in (2.3.1) can be rewritten as

$$\sum_{n=3}^{\infty} \frac{u_n}{z^{2n-1}} = \sum_{n=1}^{\infty} \frac{((n+1)! - n2^n)(2 - 2^{-n}) B_{n+1}}{n(n+1)(n+1)!} \cdot \frac{1}{z^n} =: \sum_{n=1}^{\infty} \frac{u_n^*}{z^n},$$

where

$$u_n^* = \frac{((n+1)! - n2^n)(2 - 2^{-n}) B_{n+1}}{n(n+1)(n+1)!}.$$

Clearly $u_n^* = 0$ for $1 \leq n \leq 4$. The conclusion then follows by applying the same method as in Theorem 2.2.2. $\square$

Setting $r = 1$ in the preceding theorem immediately yields the following corollary.

Corollary 2.3.3. *As $z \rightarrow \infty$ with $|\arg z| < \pi$, the asymptotic expansion*

$$\begin{aligned} \frac{\Gamma(z+1)}{\Gamma(z+1/2)} &\sim \sqrt{z} \left(\cosh \frac{1}{2z} \right)^z \left(1 + \sum_{n=1}^{\infty} \frac{v_n}{z^n} \right) \\ &= \sqrt{z} \left(\cosh \frac{1}{2z} \right)^z \left(1 + \frac{7}{5760z^5} - \frac{187}{161280z^7} + \frac{48763}{29030400z^9} + \frac{49}{66355200z^{10}} + \cdots \right) \end{aligned} \quad (2.3.5)$$

holds, where $v_0 = 1$, $v_n = 0$ for $1 \leq n \leq 4$, and for $n \geq 5$,

$$v_n = \frac{1}{n} \sum_{k=5}^n \frac{((k+1)! - k2^k)(2 - 2^{-k}) B_{k+1}}{(k+1)(k+1)!} v_{n-k}. \quad (2.3.6)$$

Next, we will give some generalized Wallis-type asymptotic expansions for the ratio of two gamma functions, the main theorem is stated as follows. Before the proof, we will give some properties of the Bernoulli polynomials and some lemmas that will be used in the proof.

Lemma 2.3.1. The Bernoulli polynomials $B_n(x)$ satisfy the following properties [14]:

- (P1) $B_n(1-x) = (-1)^n B_n(x)$;
- (P2) $B_n(x+1) - B_n(x) = nx^{n-1}$, $n = 0, 1, 2, \dots$;
- (P3) $B_n'(x) = nB_{n-1}(x)$, $n = 1, 2, \dots$;
- (P4) $(-1)^{n+1} B_{2n+1}(x) > 0$, $x \in (0, 1/2)$, $n = 1, 2, \dots$;

$$(P5) \quad B_n(1/2) = -(1 - 2^{1-n})B_n, \quad n = 0, 1, 2, \dots$$

Lemma 2.3.2. The following integral representations hold [14]:

$$\ln \Gamma(z) = \int_0^\infty \left((z-1)e^{-t} - \frac{e^{-t} - e^{-zt}}{1 - e^{-t}} \right) \frac{dt}{t}, \quad (2.3.7)$$

$$\ln z = \int_0^\infty \frac{e^{-t} - e^{-zt}}{t} dt. \quad (2.3.8)$$

Lemma 2.3.3 (Watson's Lemma). Let $\phi(t)$ be integrable over any finite interval $0 \leq t \leq T$ and infinitely differentiable at $t = 0$. Then

$$\int_0^\infty \phi(t) e^{-zt} dt \sim \sum_{n=0}^\infty \frac{\phi^{(n)}(0)}{z^{n+1}}, \quad z \rightarrow \infty. \quad (2.3.9)$$

Now comes the main results:

Theorem 2.3.4. Let $u, v \in \mathbb{R}$ with $\omega = u - v \neq 0$, and let $\sigma = (u + v - 1)/2$, $\rho = (1 - \omega)/2$. Then the following asymptotic expansion holds as $z \rightarrow \infty$ with $\operatorname{Re}(z + \sigma) > 0$, $|\arg(z + u)| < \pi$, and $|\arg(z + v)| < \pi$:

$$\left[\frac{\Gamma(z + u)}{\Gamma(z + v)} \right]^{\frac{1}{u-v}} \sim (z + \sigma) \exp \left[\sum_{n=1}^\infty \frac{B_{2n+1}(\rho)}{\omega n(2n+1)} (z + \sigma)^{-2n} \right]. \quad (2.3.10)$$

Proof. From Lemma 2.3.2, we compute

$$\begin{aligned} & \frac{\ln \Gamma(z + u) - \ln \Gamma(z + v)}{u - v} - \ln(z + \sigma) \\ &= \frac{1}{u - v} \int_0^\infty \left((z + u - 1)e^{-t} - \frac{e^{-t} - e^{-(z+u)t}}{1 - e^{-t}} \right) \frac{dt}{t} \\ & \quad - \frac{1}{u - v} \int_0^\infty \left((z + v - 1)e^{-t} - \frac{e^{-t} - e^{-(z+v)t}}{1 - e^{-t}} \right) \frac{dt}{t} - \int_0^\infty \frac{e^{-t} - e^{-(z+\sigma)t}}{t} dt \\ &= \int_0^\infty h(t) \frac{e^{-(z+\sigma)t}}{t} dt, \end{aligned}$$

where

$$h(t) = 1 - \frac{e^{(1+\omega)t/2} - e^{(1-\omega)t/2}}{\omega(e^t - 1)}.$$

Using the Bernoulli polynomial expansion and property (P1) of Lemma 2.3.1, we obtain

$$\frac{h(t)}{t} = 1 - \frac{1}{\omega t} \left[\frac{te^{(1+\omega)t/2}}{e^t - 1} - \frac{te^{(1-\omega)t/2}}{e^t - 1} \right] = \sum_{n=1}^\infty \frac{2B_{2n+1}(\rho)}{\omega(2n+1)!} t^{2n}.$$

Applying Watson's Lemma (Lemma 2.3.3) with $\phi(t) = h(t)/t$, whose $(2n-1)$ -th derivative at 0 equals $2B_{2n+1}(\rho)(2n-1)!/[\omega(2n+1)!]$, we obtain

$$\int_0^\infty \frac{h(t)}{t} e^{-(z+\sigma)t} dt \sim \sum_{n=1}^\infty \frac{2B_{2n+1}(\rho)}{\omega(2n+1)!} \cdot \frac{(2n-1)!}{(z + \sigma)^{2n}} = \sum_{n=1}^\infty \frac{2B_{2n+1}(\rho)}{\omega n(2n+1)} (z + \sigma)^{-2n},$$

which completes the proof. $\square$

Corollary 2.3.5. *Under the same hypotheses as Theorem 2.3.4, we have*

$$\left[\frac{\Gamma(z+u)}{\Gamma(z+v)} \right]^{\frac{1}{u-v}} \sim \sum_{n=0}^{\infty} \frac{c_n}{(z+\sigma)^{2n-1}}, \quad (2.3.11)$$

where $c_0 = 1$ and for $n \geq 1$,

$$c_n = \frac{1}{n} \sum_{k=1}^n k \lambda_k c_{n-k} = \frac{1}{\omega n} \sum_{k=1}^n \frac{B_{2k+1}(\rho)}{2k+1} c_{n-k}, \quad (2.3.12)$$

with $\lambda_k = B_{2k+1}(\rho)/[\omega k(2k+1)]$.

Proof. Write $A(z) = \sum_{n=1}^{\infty} \lambda_n (z+\sigma)^{-2n}$ for the exponent in (2.3.10), and set $B(z) = \exp(A(z)) \sim \sum_{n=0}^{\infty} c_n (z+\sigma)^{-2n}$ with $c_0 = 1$. Differentiating $B(z) = \exp(A(z))$ gives $B'(z) = A'(z)B(z)$, so

$$\sum_{n=0}^{\infty} c_n (-2n) (z+\sigma)^{-2n-1} = \left(\sum_{n=1}^{\infty} \lambda_n (-2n) (z+\sigma)^{-2n-1} \right) \left(\sum_{n=0}^{\infty} c_n (z+\sigma)^{-2n} \right).$$

Matching coefficients of $(z+\sigma)^{-2n-1}$ yields (2.3.12). $\square$

Corollary 2.3.6. *Under the same hypotheses as Theorem 2.3.4, we have*

$$\frac{\Gamma(z+u)}{\Gamma(z+v)} \sim \sum_{n=0}^{\infty} \frac{b_n}{(z+\sigma)^{2n-u+v}}, \quad (2.3.13)$$

where $b_0 = 1$ and for $n \geq 1$,

$$b_n = \frac{1}{n} \sum_{k=1}^n k(u-v) \lambda_k b_{n-k} = \frac{1}{\omega n} \sum_{k=1}^n \frac{B_{2k+1}(\rho)}{2k+1} b_{n-k}. \quad (2.3.14)$$

Proof. Raise both sides of (2.3.10) to the power $u-v$ and apply the same coefficient-matching argument as in Corollary 2.3.5. $\square$

Corollary 2.3.7. *Setting $u = 1$ and $v = 1/2$ in Theorem 2.3.4 gives $\omega = 1/2$, $\sigma = 1/4$, $\rho = 1/4$, and*

$$\left[\frac{\Gamma(z+1)}{\Gamma(z+1/2)} \right]^2 \sim (z + \tfrac{1}{4}) \exp \left[\sum_{n=1}^{\infty} \frac{2B_{2n+1}(1/4)}{n(2n+1)(z+1/4)^{2n}} \right], \quad z \rightarrow \infty, \quad |\arg(z+1)| < \pi. \quad (2.3.15)$$

Corollary 2.3.8. *Setting $v = 1-u$ in Theorem 2.3.4 (so that $u, v \in \mathbb{R}$, $\omega = 2u-1$, $\sigma = 0$, $\rho = 1-u$), the following asymptotic expansion holds as $z \rightarrow \infty$ with $|\arg(z+u)| < \pi$:*

$$\left[\frac{\Gamma(z+u)}{\Gamma(z+1-u)} \right]^{\frac{1}{2u-1}} \sim z \exp \left[\sum_{n=1}^{\infty} \frac{B_{2n+1}(1-u)}{(2u-1)n(2n+1)z^{2n}} \right]. \quad (2.3.16)$$

Moreover, using properties (P3) and (P5) of Lemma 2.3.1,

$$\lim_{u \rightarrow 1/2} \frac{B_{2n+1}(1-u)}{2u-1} = -\frac{1}{2} B'_{2n+1}(\tfrac{1}{2}) = -\frac{2n+1}{2} B_{2n}(\tfrac{1}{2}) = \frac{2n+1}{2} (1-2^{1-2n}) B_{2n}.$$

Taking logarithms in (2.3.16) and letting $u \rightarrow 1/2$, the left-hand side tends to $\psi(z+1/2)$, yielding the

digamma expansion

$$\psi\left(z + \frac{1}{2}\right) \sim \ln z + \sum_{n=1}^{\infty} \frac{(1 - 2^{1-2n}) B_{2n}}{2n z^{2n}}, \quad z \rightarrow \infty, \quad |\arg z| < \pi. \quad (2.3.17)$$

Next, we will give the asymptotic expansion of the Gurland-type ratio of gamma functions, which mainly concerns the ratio of the form $\frac{\Gamma(z+p)\Gamma(z+q)}{\Gamma(z+(p+q)/2)^2}$.

Theorem 2.3.9. *Let $p, q \in \mathbb{R}$, $w = p - q \neq 0$, $r_0 = (p + q - 1)/2$, $s = (1 - w)/2$. For $z \rightarrow \infty$ with $|\arg z| < \pi$, the following asymptotic expansion holds:*

$$\ln \left[\frac{\Gamma(z+p)\Gamma(z+q)}{\Gamma\left(z + \frac{p+q}{2}\right)^2} \right] \sim \sum_{n=1}^{\infty} \frac{B_{2n}(s) - B_{2n}(1/2)}{n(2n-1)(z+r_0)^{2n-1}}, \quad (2.3.18)$$

where $B_n(x)$ are the Bernoulli polynomials.

Proof. Replacing z by $z + r$ in [11], we obtain:

$$\ln \Gamma(z + r + t) \sim \left(z + r + t - \frac{1}{2}\right) \ln(z + r) - (z + r) + \frac{1}{2} \ln(2\pi) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1} B_{n+1}(t)}{n(n+1)(z+r)^n}. \quad (2.3.19)$$

Setting $t = p - r$, $t = q - r$, and $t = (p + q)/2 - r$ successively in (2.3.19) and combining, we obtain

$$\ln \frac{\Gamma(z+p)\Gamma(z+q)}{\Gamma\left(z + \frac{p+q}{2}\right)^2} \sim \sum_{k=1}^{\infty} \frac{\alpha_k}{(z+r)^k}, \quad (2.3.20)$$

where

$$\alpha_k = (-1)^{k+1} \frac{B_{k+1}(p-r) + B_{k+1}(q-r) - 2B_{k+1}\left(\frac{p+q}{2} - r\right)}{k(k+1)}. \quad (2.3.21)$$

Setting $r = r_0 = (p + q - 1)/2$, we compute

$$p - r_0 = \frac{1+w}{2} = 1-s, \quad q - r_0 = \frac{1-w}{2} = s, \quad \frac{p+q}{2} - r_0 = \frac{1}{2}. \quad (2.3.22)$$

Applying property (P1), namely $B_k(1-x) = (-1)^k B_k(x)$, we simplify α_k as

$$\alpha_k = (-1)^{k+1} \frac{B_{k+1}(1-s) + B_{k+1}(s) - 2B_{k+1}(1/2)}{k(k+1)} = \begin{cases} \frac{2B_{2n}(s) - 2B_{2n}(1/2)}{(2n)(2n-1)} & k = 2n-1, \\ 0 & k = 2n, \end{cases} \quad (2.3.23)$$

for $n \in \mathbb{N}$, where in the case $k = 2n - 1$ we used (P1) to write $B_{2n}(1-s) = B_{2n}(s)$, and in the case $k = 2n$ we used $B_{2n+1}(1-s) = -B_{2n+1}(s)$ so the two terms cancel. Substituting back into (2.3.20) yields (2.3.18). $\square$

Corollary 2.3.10. *Let $u \in \mathbb{R}$. For $z \rightarrow \infty$ with $|\arg z| < \pi$, the following asymptotic expansion holds:*

$$\ln \left[\frac{\Gamma(z)\Gamma(z+2u)}{\Gamma(z+u)^2} \right] \sim \sum_{n=1}^{\infty} \frac{B_{2n}(1/2-u) - B_{2n}(1/2)}{n(2n-1)(z+u-1/2)^{2n-1}}. \quad (2.3.24)$$

Proof. Setting $(p, q) = (2u, 0)$ in Theorem 2.3.9, we have $r_0 = u - 1/2$ and $s = 1/2 - u$, from which (2.3.24) follows immediately. $\square$

Corollary 2.3.11. *Let $p \in \mathbb{R}$. For $z \rightarrow \infty$ with $|\arg z| < \pi$, the following asymptotic expansion holds:*

$$\ln \left[\frac{\Gamma(z+p) \Gamma(z+1-p)}{\Gamma(z+1/2)^2} \right] \sim \sum_{n=1}^{\infty} \frac{B_{2n}(p) - B_{2n}(1/2)}{n(2n-1) z^{2n-1}}. \quad (2.3.25)$$

Proof. Setting $q = 1 - p$ in Theorem 2.3.9, we have $w = 2p - 1$, $r_0 = 0$, and $s = p$, from which (2.3.25) follows immediately. $\square$

Chapter 3

Recurrence relations of coefficients involving hypergeometric function in complex fields

3.1 Introduction

In this chapter, we will first give some recurrence relations of coefficients involving hypergeometric function in complex fields. And then we will give some applications of these recurrence relations in other special functions. The recurrence method is some kind of new way to deal with those complex special functions, which is different from the traditional method of using integral representations or mathematical analysis.

For complex numbers a, c , and z with $c \neq 0, -1, -2, \dots$, the confluent hypergeometric function [4] is defined as:

$$M(a; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n}{(c)_n} \frac{z^n}{n!} = {}_1F_1(a; c; z),$$

where $(a)_n$ denotes Pochhammer symbol defined by

$$(a)_n = a(a+1) \cdots (a+n-1),$$

for $n = 1, 2, \dots$, and $(a)_0 = 1$ for $a \neq 0$. Confluent hypergeometric function is a solution of a confluent hypergeometric equation, which is a degenerate form of a hypergeometric differential equation:

$$z \frac{d^2 w}{dz^2} + (c - z) \frac{dw}{dz} - aw = 0$$

where two of the three regular singularities merge into an irregular singularity. As special case, the modified Bessel function and incomplete gamma functions:

$$I_{\alpha}(x) = i^{-\alpha} J_{\alpha}(ix) = \sum_{m=0}^{\infty} \frac{1}{m! \Gamma(m + \alpha + 1)} \left(\frac{x}{2}\right)^{2m + \alpha}$$
$$\Gamma(s, z) = \int_z^{\infty} t^{s-1} e^{-t} dt$$

and

$$\gamma(s, z) = \int_0^z t^{s-1} e^{-t} dt$$

has the representations in confluent hypergeometric function:

$$I_{a-\frac{1}{2}}(z) = M(a; 2a; z) \frac{1}{\Gamma(a + \frac{1}{2})} \left(\frac{z}{4}\right)^{-\frac{1}{2}+a} e^{\frac{-z}{2}}$$

$$\Gamma(s, z) = e^{-z} U(1-s; 1-s; z)$$

and

$$\gamma(s, z) = \frac{z^s}{s} M(1; s+1; -z),$$

respectively, where $U(a; b; z) = \frac{\Gamma(1-b)}{\Gamma(a+1-b)} M(a; b; z) + \frac{\Gamma(b-1)}{\Gamma(a)} z^{1-b} M(a+1-b; 2-b; z)$.

Also, Gauss hypergeometric function [2, 3] is defined by

$$F(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!}$$

for complex numbers a, b, c, z with $c \neq 0, -1, -2, \dots$ and $|z| < 1$. Gauss hypergeometric function is a solution of Euler's hypergeometric differential equation:

$$z(1-z) \frac{d^2 w}{dz^2} + [c - (a+b+1)z] \frac{dw}{dz} - abw = 0,$$

which has three regular singular points: 0, 1 and ∞ . The complete elliptic integrals of the first and second kinds,

$$K(r) = \int_0^{\pi/2} \frac{dt}{\sqrt{1-r^2 \sin^2 t}} \quad \text{and} \quad E(r) = \int_0^{\pi/2} \sqrt{1-r^2 \sin^2 t} dt,$$

could be represented respectively by the hypergeometric function representations

$$K(r) = \frac{\pi}{2} F\left(\frac{1}{2}, \frac{1}{2}; 1; r^2\right) \quad \text{and} \quad E(r) = \frac{\pi}{2} F\left(-\frac{1}{2}, \frac{1}{2}; 1; r^2\right).$$

In the study of certain special functions, one often encounters the functions $\mathcal{M}(z)$ and $\mathcal{G}(z)$. If p is an integer, then it is easy to deal with $\mathcal{M}(z)$ and $\mathcal{G}(z)$. If p is not an integer and $p \neq a+b-c$, then $\mathcal{M}(z)$ and $\mathcal{G}(z)$ are difficult to be treated. Then, how to deal with this kind of problem?

As is known to all, giving recurrence relations is one of important ways to study some special functions. In 2018, Yang and Tian [18] offered two recurrence relations of Maclaurin series coefficients of $(r')^p K(r)$ and $(r')^p E(r)$, $r, p \in \mathbb{R}$. In 2021, Yang and Tian [16] presented another recurrence relation of Maclaurin series coefficients of $(1-x)^{-q} F(-\frac{1}{2}, -\frac{1}{2}; 2; x)$, $x, q \in \mathbb{R}$, and provided the sharp lower and upper bounds for $E(r)$ in terms of weighted power means of arithmetic and geometric means. In 2025, Yang [19] provided the recurrence relation of the $(1-\theta x)^p F(a, b; c; x)$, $\theta, p, a, b, c, x \in \mathbb{R}$.

Inspired by the recurrence formulas established in [16] and [19], the main aim of this paper is to establish a general recurrence relation of $\mathcal{M}(z)$ and $\mathcal{G}(z)$ ' Maclaurin series coefficients in complex fields.

3.2 Recurrence relations of coefficients involving confluent hypergeometric function

We start with a necessary lemma for the confluent hypergeometric function.

Lemma 3.2.1. For, $a, b, z \in \mathbb{C}$, $-b \notin \mathbb{N} \cup \{0\}$, let

$$M \equiv M(a; c; z), \quad M(a-) \equiv M(a-1; c; z), \quad M(a+) \equiv M(a+1; c; z).$$

we have the following relation:

$$z \frac{dM}{dz} = a(M(a+) - M) = (c-a)M(a-) + (a-c+z)M,$$

while the first equation could also be written as:

$$z \frac{dM(a-)}{dz} = (a-1)(M(a) - M(a-))$$

Theorem 3.2.1. Let $a, c, p, \theta, z \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$, and let $|\arg(1-\theta z)| < \pi$. Suppose that

$$\mathcal{M}(z) = (1-\theta z)^p M(a, c; z) = \sum_{n=0}^{\infty} u_n z^n, \quad |z| < \frac{1}{|\theta|}.$$

Then $u_0 = 1$, $u_1 = \frac{a}{c} - \theta p$, $u_2 = \frac{1}{2} \left(\frac{a^2+a}{c^2+c} - \frac{2a\theta p}{c} + \theta^2(p-1)p \right)$ and

$$u_{n+1} = \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2},$$

where

$$\begin{aligned} \beta_0(n) &= \frac{a + 2\theta n(c+n-1) - \theta p(c+2n) + n}{(n+1)(c+n)}, \\ \beta_1(n) &= \frac{\theta(-2a - \theta(n-p-1)(c+n-p-2) - 2n + p + 2)}{(n+1)(c+n)}, \\ \beta_2(n) &= \frac{\theta^2(a+n-p-2)}{(n+1)(c+n)}. \end{aligned}$$

Proof. Let $\mathcal{M}^*(z) = (1-\theta z)^p M(a-1; c; z) = (1-\theta z)^p M(a-) = \sum_{n=0}^{\infty} u'_n z^n$. We first differentiate both two power series expressions and multiply $z(1-\theta z)$, then we get

$$-p\theta z \mathcal{M}(z) + (1-\theta z)((c-a)\mathcal{M}^*(z) + (a-c+z)\mathcal{M}(z)) = (z-\theta z^2) \sum_{n=0}^{\infty} n u_n z^{n-1},$$

and

$$-p\theta z \mathcal{M}^*(z) + (a-1)(1-\theta z)(\mathcal{M}(z) - \mathcal{M}^*(z)) = (z-\theta z^2) \sum_{n=0}^{\infty} n u'_n z^{n-1}.$$

Then, we just expand $\mathcal{M}$ and $\mathcal{M}^*$ into power series:

$$\begin{aligned}
& (b-a)u'_0 + (a-c)u_0 + z[-p\theta u_0 + (c-a)u'_1 - \theta(b-c)u'_0 + (a-c)u_1 + (1-\theta(a-c))u_0] + \\
& \sum_{n=2}^{\infty} z^n (-p\theta u_{n-1} + (c-a)u'_n - \theta(c-a)u'_{n-1} + (a-c)u_n + (1-\theta(a-c))u_{n-1} - \theta u_{n-2}) \\
& = u_1 z + \sum_{n=0}^{\infty} (nu_n - \theta u_{n-1}) z^n
\end{aligned}$$

and

$$\begin{aligned}
& (a-1)(u_0 - u'_0) + z(-p\theta u'_0 + (a-1)u_1 - (a-1)\theta u_0 - (a-1)u'_1 + (a-1)\theta u'_0) + \\
& \sum_{n=2}^{\infty} z^n (-p\theta u'_{n-1} + (a-1)(u_n - u'_n) - \theta(a-1)(u_{n-1} - u'_{n-1})) \\
& = zu'_1 + \sum_{n=2}^{\infty} z^n (nu'_n - (n-1)u'_{n-1}).
\end{aligned}$$

Compare the coefficients of z^n between these equations, then we get

$$\begin{aligned}
u_0 &= 1 \\
u_1 &= \frac{a}{c} - \theta p \\
(c-a)(u'_n - u'_{n-1}) &= (1+c-a)u_n + (\theta(a-c+1) + p\theta - 1)u_{n-1} + \theta u_{n-2} \\
(a-1)(u_n - u_{n-1}) &= au'_n + (p\theta - \theta a)u'_{n-1}
\end{aligned}$$

Represent the u'_{n-1} by u_n , u_{n-1} and u_{n-2} , we obtain

$$u'_{n-1} = \frac{C_0 u_n + C_1 u_{n-1} + C_2 u_{n-2}}{\theta(a-c)(p+1)},$$

where, $C_0 = n(c+n-1)$, $C_1 = \theta(ap + a - cn - n^2 + np + 2n - p - 1) - a - n + 1$,

$$C_2 = \theta(a+n-1).$$

Now we substitute the recurrence relation of u'_n into the original equation, and finally get the recurrence relation of u_n . While for u_2 , we just consider the Taylor expansion of $(1-\theta z)^p$ and $M(a; c; z)$:

$$\begin{aligned}
(1-\theta z)^p &= 1 - p\theta + \frac{p(p-1)\theta}{2} z^2 + o(z^3) \\
M(a; c; z) &= 1 + \frac{a}{c} z + \frac{a(a+1)}{c(c+1)} z^2 + o(z^3)
\end{aligned}$$

It is easy to see that the coefficient of z^2 is $u_2 = \frac{1}{2} \left(\frac{a^2+a}{c^2+c} - \frac{2a\theta p}{c} + \theta^2(p-1)p \right)$, which completes the proof. $\square$

If we restrict $p \in \mathbb{Z}$, we would get the following result from Theorem 3.2.1 without setting $\arg(1-\theta z) \in (-\pi, \pi)$ canonically.

Theorem 3.2.2. *Let $a, c, \theta, z \in \mathbb{C}$, $p \in \mathbb{Z}$, $-c \notin \mathbb{N} \cup \{0\}$. Then*

$$(1-\theta z)^p M(a; c; z) = \sum_{n=0}^{\infty} u_n z^n, \quad |z| < \frac{1}{|\theta|},$$

with $u_0 = 1$, $u_1 = \frac{a}{c} - \theta p$, $u_2 = \frac{1}{2} \left(\frac{a^2+a}{c^2+c} - \frac{2a\theta p}{c} + \theta^2(p-1)p \right)$ and

$$u_{n+1} = \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2},$$

where

$$\begin{aligned}\beta_0(n) &= \frac{a + 2\theta n(c+n-1) - \theta p(c+2n) + n}{(n+1)(c+n)}, \\ \beta_1(n) &= \frac{\theta(-2a - \theta(n-p-1)(c+n-p-2) - 2n+p+2)}{(n+1)(c+n)}, \\ \beta_2(n) &= \frac{\theta^2(a+n-p-2)}{(n+1)(c+n)}.\end{aligned}$$

Now we will give some special case of Theorem 3.2.1 for $\theta = 1, -1, i, 1+i$. Substitute $\theta = 1$ into Theorem 3.2.1, we could get the following result.

Corollary 3.2.3. *Let $a, c, z, p \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$, and let $|\arg(1-z)| < \pi$. Then*

$$(1-z)^p M(a, c; z) = \sum_{n=0}^{\infty} u_n z^n, \quad |z| < 1,$$

with $u_0 = 1$, $u_1 = \frac{a}{c} - p$, $u_2 = \frac{1}{2} \left(\frac{a^2+a}{c^2+c} - \frac{2ap}{c} + (p-1)p \right)$ and

$$u_{n+1} = \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2},$$

where

$$\begin{aligned}\beta_0(n) &= \frac{a + 2n(c+n-1) - p(c+2n) + n}{(n+1)(c+n)}, \\ \beta_1(n) &= \frac{(-2a - (n-p-1)(c+n-p-2) - 2n+p+2)}{(n+1)(c+n)}, \\ \beta_2(n) &= \frac{(a+n-p-2)}{(n+1)(c+n)}.\end{aligned}$$

Substitute $\theta = -1$ into Theorem 3.2.1, we could get the following result.

Corollary 3.2.4. *Let $a, c, z, p \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$, and let $|\arg(1+z)| < \pi$. Then*

$$(1+z)^p M(a, c; z) = \sum_{n=0}^{\infty} u_n z^n, \quad |z| < 1,$$

with $u_0 = 1$, $u_1 = \frac{a}{c} + p$, $u_2 = \frac{1}{2} \left(\frac{a^2+a}{c^2+c} + \frac{2ap}{c} + (p-1)p \right)$ and

$$u_{n+1} = \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2},$$

where

$$\begin{aligned}\beta_0(n) &= \frac{a - 2n(c+n-1) + p(c+2n) + n}{(n+1)(c+n)}, \\ \beta_1(n) &= \frac{(2a - (n-p-1)(c+n-p-2) + 2n-p-2)}{(n+1)(c+n)}, \\ \beta_2(n) &= \frac{(a+n-p-2)}{(n+1)(c+n)}.\end{aligned}$$

Substitute $\theta = i$ into Theorem 3.2.1, then we have the following result.

Corollary 3.2.5. *Let $a, c, z, p \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$, and let $|\arg(1 - iz)| < \pi$. Then*

$$(1 - iz)^p M(a, c; z) = \sum_{n=0}^{\infty} u_n z^n, \quad |z| < 1,$$

with $u_0 = 1$, $u_1 = \frac{a}{c} - ip$, $u_2 = \frac{1}{2} \left(\frac{a^2 + a}{c^2 + c} - \frac{2iap}{c} - (p - 1)p \right)$ and

$$u_{n+1} = \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2},$$

where

$$\begin{aligned} \beta_0(n) &= \frac{a + 2in(c + n - 1) - ip(c + 2n) + n}{(n + 1)(c + n)}, \\ \beta_1(n) &= \frac{i(-2a - i(n - p - 1)(c + n - p - 2) - 2n + p + 2)}{(n + 1)(c + n)}, \\ \beta_2(n) &= -\frac{(a + n - p - 2)}{(n + 1)(c + n)}. \end{aligned}$$

Substitute $\theta = 1 + i$ into Theorem 3.2.1, we could get the following result.

Corollary 3.2.6. *Let $a, c, z, p \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$, and let $|\arg(1 - (1 + i)z)| < \pi$. Then*

$$(1 - (1 + i)z)^p M(a, c; z) = \sum_{n=0}^{\infty} u_n z^n, \quad |z| < \frac{1}{\sqrt{2}},$$

with $u_0 = 1$, $u_1 = \frac{a}{c} - (1 + i)p$, $u_2 = \frac{1}{2} \left(\frac{a^2 + a}{c^2 + c} - \frac{2a(1+i)p}{c} + 2i(p - 1)p \right)$ and

$$u_{n+1} = \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2},$$

where

$$\begin{aligned} \beta_0(n) &= \frac{a + 2(1 + i)n(c + n - 1) - (1 + i)p(c + 2n) + n}{(n + 1)(c + n)}, \\ \beta_1(n) &= \frac{(1 + i)(-2a - (1 + i)(n - p - 1)(c + n - p - 2) - 2n + p + 2)}{(n + 1)(c + n)}, \\ \beta_2(n) &= \frac{2i(a + n - p - 2)}{(n + 1)(c + n)}. \end{aligned}$$

3.3 Recurrence relations of coefficients involving gauss hypergeometric function

We start with a useful lemma for Gauss hypergeometric function.

Lemma 3.3.1. [19] Let $a, b, c, z \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$ and $|z| < 1$

$$F \equiv F(a, b; c, z), \quad F_{a-} \equiv F(a - 1, b; c, z), \quad F_{a+} \equiv F(a + 1, b; c, z)$$

Then

$$\frac{dF}{dz} = \frac{(c-a)F_{a-} + (a-c+bz)F}{z(1-z)}, \quad (3.3.1)$$

$$\frac{dF_{a-}}{dz} = \frac{a-1}{z}(F-F_{a-}) \quad (3.3.2)$$

We are now going to present and prove the main theorem for Gauss hypergeometric function.

Theorem 3.3.1. *Let $a, b, c, p, \theta, z \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$. Then we have*

$$\mathcal{G}(z) = (1-\theta z)^p F(a, b; c; z) = \sum_{n=0}^{\infty} v_n z^n,$$

with $|z| < 1$, $|\arg(1-\theta z)| < \pi$, $v_0 = 1$, $v_1 = ab/c - p\theta$,

$$v_2 = \frac{1}{2}\theta^2 p(p-1) - \theta p \frac{ab}{c} + \frac{1}{2} \frac{ab(b+1)(a+1)}{c(c+1)},$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p,\theta}(a, b, c)}{(n+1)(n+c)} v_n - \theta \frac{\zeta_{n,p,\theta}(a, b, c)}{(n+1)(n+c)} v_{n-1} + \theta^2 \frac{\sigma_{n,p}(a, b)}{(n+1)(n+c)} v_{n-2}, \quad (3.3.3)$$

where

$$\begin{aligned} \mu_{n,p,\theta}(a, b, c) &= (n+a)(n+b) + \theta [2n^2 - 2n(p-c+1) - cp], \\ \zeta_{n,p,\theta}(a, b, c) &= 2n^2 + 2(a+b-p-2)n - (a+b-1)p + 2(a-1)(b-1) \\ &\quad + \theta(n-p-1)(n-p+c-2), \\ \sigma_{n,p}(a, b) &= (n+a-p-2)(n+b-p-2). \end{aligned}$$

Proof. Denote

$$\mathcal{G}^*(z) \equiv (1-\theta z)^p F(a-1, b; c; z) = \sum_{n=0}^{\infty} v_n^* z^n.$$

The proof is analogous to that of Theorem 3.2.1 in Section 2, we first consider differentiating $\mathcal{G}(z)$

$$-p\theta(1-\theta z)^{p-1}F + (1-\theta z)^p F' = \sum_{n=0}^{\infty} n v_n z^{n-1}.$$

Substituting (3.3.1) into the above equation and multiplying by $z(1-z)(1-\theta z)$ give

$$\begin{aligned} -p\theta z(1-z)\mathcal{G}(z) + (c-a)(1-\theta z)\mathcal{G}^*(z) + (a-c+bz)(1-\theta z)\mathcal{G}(z) \\ = (1-z)(1-\theta z) \sum_{n=0}^{\infty} n v_n z^n. \end{aligned} \quad (3.3.4)$$

Now we consider expanding $\mathcal{G}$ and $\mathcal{G}^*$ into the assumed power series:

$$\begin{aligned} &-p\theta \left(\sum_{n=1}^{\infty} v_{n-1} z^n - \sum_{n=2}^{\infty} v_{n-2} z^n \right) + (c-a) \left(\sum_{n=0}^{\infty} v_n^* z^n - \theta \sum_{n=1}^{\infty} v_{n-1}^* z^n \right) \\ &+ (a-c) \sum_{n=0}^{\infty} v_n z^n + (b-\theta(a-c)) \sum_{n=1}^{\infty} v_{n-1} z^n - b\theta \sum_{n=2}^{\infty} v_{n-2} z^n \\ &= \sum_{n=1}^{\infty} n v_n z^n - (\theta+1) \sum_{n=2}^{\infty} (n-1) v_{n-1} z^n + \theta \sum_{n=2}^{\infty} (n-2) v_{n-2} z^n. \end{aligned}$$

We may rewrite the above equation like

$$\begin{aligned}
& (c-a)(v_0^* - v_0) + [-p\theta v_0 + (c-a)(v_1^* - \theta v_0^*) + (a-c)v_1 + (b-\theta(a-c))v_0]z \\
& + \sum_{n=2}^{\infty} [(c-a)(v_n^* - \theta v_{n-1}^*) + (a-c)v_n + (b-\theta(a-c+p))v_{n-1} + (p-b)\theta v_{n-2}]z^n \\
& = v_1 z + \sum_{n=2}^{\infty} [nv_n - (\theta+1)(n-1)v_{n-1} + \theta(n-2)v_{n-2}]z^n.
\end{aligned}$$

Then, we could give the following results by comparing coefficients of z^n ,

$$\begin{aligned}
(c-a)(v_0^* - v_0) &= 0, \quad (c-a)(v_1^* - \theta v_0^*) + (b-\theta(a-c+p))v_0 + (a-c)v_1 = v_1, \\
(c-a)(v_n^* - \theta v_{n-1}^*) &= (n-a+c)v_n + \theta(n-2+b-p)v_{n-2} \\
&+ [\theta(a-c+p) - b - (\theta+1)(n-1)]v_{n-1} \text{ for } n \geq 2.
\end{aligned} \tag{3.3.5}$$

In the same way, we will differentiate $\mathcal{G}^*$ and get

$$-p\theta(1-\theta z)^{p-1}F_{a-} + (1-\theta z)^p F'_{a-} = \sum_{n=0}^{\infty} nv_n^* z^{n-1}.$$

Substituting (3.3.2) into the above equation and multiplying by $x(1-\theta x)$ give:

$$-p\theta z\mathcal{G}^*(z) + (a-1)(1-\theta z)(\mathcal{G}(z) - \mathcal{G}^*(z)) = (1-\theta z) \sum_{n=0}^{\infty} nv_n^* z^n.$$

Then, we still consider expanding $\mathcal{G}$ and $\mathcal{G}^*$ in power series and get

$$-p\theta \sum_{n=1}^{\infty} v_{n-1}^* z^n + (a-1)(1-\theta z) \sum_{n=0}^{\infty} (v_n - v_n^*) z^n = (1-\theta z) \sum_{n=0}^{\infty} nv_n^* z^n,$$

which can be rewritten as

$$\begin{aligned}
& (a-1)(v_0 - v_0^*) - p\theta \sum_{n=1}^{\infty} v_{n-1}^* z^n + (a-1) \sum_{n=1}^{\infty} (v_n - \theta v_{n-1} - v_n^* + \theta v_{n-1}^*) z^n \\
& = \sum_{n=1}^{\infty} (nv_n^* - \theta(n-1)v_{n-1}^*) z^n.
\end{aligned}$$

From the equality of the coefficients of z^n , it follows that $(a-1)(v_0 - v_0^*) = 0$ and

$$(n+a-1)v_n^* - \theta(n-2+a-p)v_{n-1}^* = (a-1)v_n - \theta(a-1)v_{n-1} \text{ for } n \geq 1. \tag{3.3.6}$$

Now from equations (3.3.5) and (3.3.6), we obtain the representation for v_{n-1}^* by v_n, v_{n-1}, v_{n-2} with $n \geq 2$

$$\begin{aligned}
& \theta(p+1)(a-c)v_{n-1}^* = n(n+c-1)v_n + \theta(n+a-1)(n+b-p-2)v_{n-2} \\
& - [(\theta+1)n^2 + (a+b+(c-p-2)\theta-2)n + (a-1)(b-(p+1)\theta-1)]v_{n-1}.
\end{aligned} \tag{3.3.7}$$

When $\theta(p+1) \neq 0$, substituting the above expressions of v_{n-1}^* and v_n^* into (3.3.5) gives

$$\begin{aligned} & v_{n+1} - \frac{(2\theta+1)n^2 + (a+b+2\theta(c-1-p))n + (ab-cp\theta)}{(n+1)(n+c)}v_n \\ & + \theta \frac{(\theta+2)n^2 + (2a+2b-2p+\theta(c-2p-3)-4)n + \theta(p+1)(p-c+2) + 2ab - (p+2)(a+b-1)}{(n+1)(n+c)}v_{n-1} \\ & - \theta^2 \frac{(n+a-p-2)(n+b-p-2)}{(n+1)(n+c)}v_{n-2} = 0 \text{ for } n \geq 2, \end{aligned}$$

which implies (3.3.3).

Also, we have the following expression by using the Cauchy product formula

$$v_n = \sum_{k=0}^n \frac{(a)_k (b)_k}{k! (c)_k} \frac{\theta^{n-k} (-p)_{n-k}}{(n-k)!},$$

which means $v_0 = 1$,

$$v_1 = \frac{ab}{c} - p\theta \quad \text{and} \quad v_2 = \frac{1}{2}\theta^2 p(p-1) - \theta p \frac{ab}{c} + \frac{1}{2} \frac{ab(b+1)(a+1)}{c(c+1)}.$$

When $\theta = 0$, the recurrence relation (3.3.3) holds trivially. When $p = -1$, it could be seen that

$$v_n = \sum_{k=0}^n \frac{(a)_k (b)_k}{k! (c)_k} \theta^{n-k},$$

which could be written as

$$v_{n+1} - \theta v_n = \frac{(a)_{n+1} (b)_{n+1}}{(n+1)! (c)_{n+1}}.$$

This also satisfied the recurrence relation (3.3.3), and the whole proof is done. $\square$

We have the following result by substituting $\theta = -1$ into Theorem 3.3.1.

Corollary 3.3.2. *Let $a, b, c, p, z \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$, $|z| < 1$, $|\arg(1+z)| < \pi$. Then*

$$(1+z)^p F(a, b; c; z) = \sum_{n=0}^{\infty} v_n z^n,$$

with $v_0 = 1$, $v_1 = ab/c + p$,

$$v_2 = \frac{1}{2}p(p-1) + p \frac{ab}{c} + \frac{1}{2} \frac{ab(b+1)(a+1)}{c(c+1)},$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p}(a, b, c)}{(n+1)(n+c)}v_n + \frac{\zeta_{n,p}(a, b, c)}{(n+1)(n+c)}v_{n-1} + \frac{\sigma_{n,p}(a, b)}{(n+1)(n+c)}v_{n-2},$$

where

$$\begin{aligned} \mu_{n,p}(a, b, c) &= -n^2 + (a+b-2c+2p+2)n + (ab+cp), \\ \zeta_{n,p}(a, b, c) &= (n+2a+2b-c)(n-1) - p^2 - (a+b-c+2)p + 2ab, \\ \sigma_{n,p}(a, b) &= (n+a-p-2)(n+b-p-2). \end{aligned}$$

Substituting $\theta = 1$ into Theorem 3.3.1, we can easily obtain a special 3-order recurrence relation. However, the order can be reduced from 3 to 2 from the proof of Theorem 3.3.1.

Corollary 3.3.3. *Let For $a, b, c, p, z \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$. Then*

$$\mathcal{G}(z) = (1-z)^p F(a, b; c; z) = \sum_{n=0}^{\infty} u_n z^n,$$

with $|z| < 1$, $|\arg(1-z)| < \pi$, $v_0 = 1$, $v_1 = ab/c - p$ and for $n \geq 1$,

$$v_{n+1} = \tau_n v_n - \rho_n v_{n-1}, \quad (3.3.8)$$

where

$$\tau_n = \frac{2n^2 + (a+b+c-2p-1)n + ab - cp}{(n+1)(n+c)}, \quad (3.3.9)$$

$$\rho_n = \frac{(n+a-p-1)(n+b-p-1)}{(n+1)(n+c)}. \quad (3.3.10)$$

Proof. When $\theta = 1$, dividing $(1-z)$ by the relation (3.3.4) gives

$$-pz\mathcal{G}(z) + (c-a)\mathcal{G}^*(z) + (a-c+bz)\mathcal{G}(z) = (1-z) \sum_{n=0}^{\infty} nv_n z^n.$$

Expanding $\mathcal{G}(z)$ and $\mathcal{G}^*(z)$ in power series gives

$$\begin{aligned} & -p \sum_{n=1}^{\infty} v_{n-1} z^n + (c-a) \sum_{n=0}^{\infty} v_n^* z^n + (a-c) \sum_{n=0}^{\infty} v_n z^n + b \sum_{n=1}^{\infty} v_{n-1} z^n \\ &= \sum_{n=1}^{\infty} nv_n z^n - \sum_{n=1}^{\infty} (n-1) v_{n-1} z^n, \end{aligned}$$

Comparing coefficients of z^n gives $(c-a)(v_0^* - v_0) = 0$,

$$-pv_{n-1} + (c-a)v_n^* + (a-c)v_n + bv_{n-1} = nv_n - (n-1)v_{n-1} \text{ for } n \geq 1,$$

which implies

$$(c-a)v_n^* = (n-a+c)v_n - (n-1+b-p)v_{n-1} \text{ for } n \geq 1. \quad (3.3.11)$$

When $c \neq a$, eliminating v_n^* and v_{n-1}^* from the Eqs. (3.3.11) and (3.3.6) for $\theta = 1$ yields

$$\begin{aligned} v_n &= \frac{2(n-1)^2 + (a+b+c-2p-1)(n-1) + ab - cp}{n(c+n-1)} v_{n-1} \\ &\quad - \frac{(n+a-p-2)(n+b-p-2)}{n(c+n-1)} v_{n-2} \text{ for } n \geq 2. \end{aligned}$$

Replacing n by $n+1$ give the desired recurrence formula (3.3.8).

When $c = a$, from the relation (3.3.11) we have

$$v_n = \frac{n-1+b-p}{n} v_{n-1} \text{ for } n \geq 1,$$

which, by an easy check, satisfies the recurrence relation (3.3.8).

The values of v_0 and v_1 follows Theorem 3.3.1, and the proof is done. $\square$

We have the following result by substituting $\theta = -i$ into Theorem 3.3.1.

Corollary 3.3.4. *Let $a, b, c, p, z \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$. Then*

$$(1 - iz)^p F(a, b; c; z) = \sum_{n=0}^{\infty} u_n z^n,$$

with $|z| < 1$, $|\arg(1 - iz)| < \pi$, $v_0 = 1$, $v_1 = ab/c - ip$,

$$v_2 = -\frac{1}{2}p(p-1) - ip\frac{ab}{c} + \frac{1}{2}\frac{ab(b+1)(a+1)}{c(c+1)},$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p}(a, b, c)}{(n+1)(n+c)}v_n - i\frac{\zeta_{n,p}(a, b, c)}{(n+1)(n+c)}v_{n-1} - \frac{\sigma_{n,p}(a, b)}{(n+1)(n+c)}v_{n-2},$$

where

$$\begin{aligned}\mu_{n,p}(a, b, c) &= (n+a)(n+b) + i[2n^2 - 2n(p-c+1) - cp], \\ \zeta_{n,p}(a, b, c) &= 2n^2 + 2(a+b-p-2)n - (a+b-1)p + 2(a-1)(b-1) \\ &\quad + i(n-p-1)(n-p+c-2), \\ \sigma_{n,p}(a, b) &= (n+a-p-2)(n+b-p-2).\end{aligned}$$

We have the following result by substituting $\theta = 1 + i$ into Theorem 3.3.1.

Corollary 3.3.5. *Let $a, b, c, p, z \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$. Then*

$$(1 - (1+i)z)^p F(a, b; c; z) = \sum_{n=0}^{\infty} u_n z^n,$$

with $|z| < 1$, $z \neq \frac{1}{1+i}$, $|\arg(1 - (1+i)z)| < \pi$, $v_0 = 1$, $v_1 = ab/c - (1+i)p$,

$$v_2 = -\frac{1}{2}p(p-1) - (1+i)p\frac{ab}{c} + \frac{1}{2}\frac{ab(b+1)(a+1)}{c(c+1)},$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p}(a, b, c)}{(n+1)(n+c)}v_n - (1+i)\frac{\zeta_{n,p}(a, b, c)}{(n+1)(n+c)}v_{n-1} + 2i\frac{\sigma_{n,p}(a, b)}{(n+1)(n+c)}v_{n-2},$$

where

$$\begin{aligned}\mu_{n,p}(a, b, c) &= (n+a)(n+b) + (1+i)[2n^2 - 2n(p-c+1) - cp], \\ \zeta_{n,p}(a, b, c) &= 2n^2 + 2(a+b-p-2)n - (a+b-1)p + 2(a-1)(b-1) \\ &\quad + (1+i)(n-p-1)(n-p+c-2), \\ \sigma_{n,p}(a, b) &= (n+a-p-2)(n+b-p-2).\end{aligned}$$

3.4 Applications

We are now going to use the main results in Section 2 and 3 to give some Maclaurin recurrence relations involving other special functions.

3.4.1 Applications of the confluent hypergeometric function

First, we present a recurrence relation involving the error function $Erf(z)$.

Corollary 3.4.1. *For $z, p \in \mathbb{C}$, $|z|^2 < \frac{1}{|\theta|}$ and $\arg(1 + \theta z^2) \in (-\pi, \pi)$, we could have a recurrence relation of coefficients for the error function:*

$$(1 + \theta z^2)^p Erf(z) = \sum_{n=0}^{\infty} a_n z^{2n+1}$$

with $a_n = \frac{(-1)^n u_n}{\sqrt{\pi}}$, $u_0 = 1$, $u_1 = \frac{1}{3} - \theta p$, $u_2 = \frac{1}{2} \left(\frac{1}{5} - \frac{2\theta p}{3} + \theta^2(p-1)p \right)$ and

$$u_{n+1} = \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2},$$

where

$$\begin{aligned} \beta_0(n) &= \frac{1 + 2\theta n(1 + 2n) - \theta p(3 + 4n) + n}{(n+1)(3 + 2n)}, \\ \beta_1(n) &= \frac{\theta(-1 - \theta(n-p-1)(\frac{3}{2} + n - p - 2) - 2n + p + 2)}{(n+1)(\frac{3}{2} + n)}, \\ \beta_2(n) &= \frac{\theta^2(2n - 2p - 3)}{(n+1)(3 + 2n)}, \end{aligned}$$

Proof. It is known all that:

$$Erf(z) = \frac{z}{\sqrt{\pi}} M\left(\frac{1}{2}; \frac{3}{2}; -z^2\right)$$

Then, applying Theorem 3.2.1, we have:

$$(1 + \theta z^2)^p Erf(z) = \frac{z}{\sqrt{\pi}} (1 + \theta z^2)^p M\left(\frac{1}{2}; \frac{3}{2}; -z^2\right) = \frac{z}{\sqrt{\pi}} \sum_{n=0}^{\infty} u_n (-z^2)^n,$$

with $u_0 = 1$, $u_1 = \frac{1}{3} - \theta p$, $u_2 = \frac{1}{2} \left(\frac{1}{5} - \frac{2\theta p}{3} + \theta^2(p-1)p \right)$ and

$$u_{n+1} = \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2},$$

where

$$\begin{aligned} \beta_0(n) &= \frac{1 + 2\theta n(1 + 2n) - \theta p(3 + 4n) + n}{(n+1)(3 + 2n)}, \\ \beta_1(n) &= \frac{\theta(-1 - \theta(n-p-1)(\frac{3}{2} + n - p - 2) - 2n + p + 2)}{(n+1)(\frac{3}{2} + n)}, \\ \beta_2(n) &= \frac{\theta^2(2n - 2p - 3)}{(n+1)(3 + 2n)}, \end{aligned}$$

which could be written as:

$$(1 + \theta z^2)^p Erf(z) = \sum_{n=0}^{\infty} a_n z^{2n+1},$$

with $a_n = \frac{(-1)^n u_n}{\sqrt{\pi}}$, which completes the proof. □

Next, we present recurrence relations involving the lower(upper) incomplete gamma function $\gamma(s, z)$

$(\Gamma(s, z))$ respectively.

Corollary 3.4.2. *Let $s, z \in \mathbb{C}$, $-(s+1) \notin \mathbb{N} \cup \{0\}$, then we have:*

$$(1 - \theta z)^p \gamma(s, z) = (1 - \theta z)^p \frac{z^s}{s} M(1; s+1; -z) = \sum_{n=0}^{\infty} \frac{(-1)^{n+s} u_n}{s} z^{n+s}$$

with $z \in \mathbb{C}$, $|z| < \frac{1}{|\theta|}$, $|\arg(1 - \theta z)| < \pi$ and $|\arg z| < \pi$, $u_0 = 1$, $u_1 = \frac{1}{s+1} - \theta p$, $u_2 = \frac{1}{2} \left(\frac{2}{(s+1)^2 + s+1} - \frac{2\theta p}{s+1} + \theta^2(p-1)p \right)$ and

$$u_{n+1} = \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2},$$

where

$$\begin{aligned} \beta_0(n) &= \frac{1 + 2\theta n(s+n) - \theta p(s+1+2n) + n}{(n+1)(s+1+n)}, \\ \beta_1(n) &= \frac{\theta(-2 - \theta(n-p-1)(s+n-p-1) - 2n+p+2)}{(n+1)(s+1+n)}, \\ \beta_2(n) &= \frac{\theta^2(n-p-1)}{(n+1)(s+1+n)}. \end{aligned}$$

Proof. Applying Theorem 3.2.1, we just get:

$$(1 - \theta z)^p M(1; s+1; -z) = \sum_{n=0}^{\infty} u_n (-z)^n,$$

with $u_0 = 1$, $u_1 = \frac{1}{s+1} - \theta p$, $u_2 = \frac{1}{2} \left(\frac{2}{(s+1)^2 + s+1} - \frac{2\theta p}{s+1} + \theta^2(p-1)p \right)$ and

$$u_{n+1} = \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2},$$

where

$$\begin{aligned} \beta_0(n) &= \frac{1 + 2\theta n(s+n) - \theta p(s+1+2n) + n}{(n+1)(s+1+n)}, \\ \beta_1(n) &= \frac{\theta(-2 - \theta(n-p-1)(s+n-p-1) - 2n+p+2)}{(n+1)(s+1+n)}, \\ \beta_2(n) &= \frac{\theta^2(n-p-1)}{(n+1)(s+1+n)}. \end{aligned}$$

Then,

$$(1 - \theta z)^p \gamma(s, z) = \sum_{n=0}^{\infty} \frac{u_n}{s} (-z)^{n+s}$$

which completes the proof. $\square$

Corollary 3.4.3. *Let $s, z, p \in \mathbb{C}$, $|z| < \frac{1}{|\theta|}$ with $\arg(1 - \theta z)$ and $\arg z \in (-\pi, \pi)$, then we have:*

$$(1 - \theta z)^p \Gamma(s, z) = \sum_{n=0}^{\infty} v_n z^n + \sum_{n=0}^{\infty} w_n z^{n+s},$$

with $v_n = \Gamma(s) \sum_{k=0}^n \frac{(-1)^k u_{n-k}}{k!}$, $w_n = \frac{-1}{s} \sum_{k=0}^n \frac{(-1)^k u'_{n-k}}{k!}$, $u_0 = 1$, $u_1 = 1 - \theta p$, $u_2 = \frac{1}{2} (1 - 2\theta p + \theta^2(p-1)p)$,

$$u'_0 = 1, u'_1 = \frac{1}{s+1} - \theta p, u'_2 = \frac{1}{2} \left(\frac{2}{(s+1)(s+2)} - \frac{2\theta p}{s+1} + \theta^2(p-1)p \right) \text{ and}$$

$$u_{n+1} = \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2},$$

$$u'_{n+1} = \beta'_0(n)u'_n + \beta'_1(n)u'_{n-1} + \beta'_2(n)u'_{n-2},$$

where

$$\begin{aligned}\beta_0(n) &= \frac{1-s+2\theta n(n-s)-\theta p(1-s+2n)+n}{(n+1)(1-s+n)}, \\ \beta_1(n) &= \frac{\theta(-2(1-s)-\theta(n-p-1)(n-s-p-1)-2n+p+2)}{(n+1)(1-s+n)}, \\ \beta_2(n) &= \frac{\theta^2(n-s-p-1)}{(n+1)(1-s+n)}.\end{aligned}$$

$$\begin{aligned}\beta'_0(n) &= \frac{1+2\theta n(c+n-1)-\theta p(s+1+2n)+n}{(n+1)(s+1+n)}, \\ \beta'_1(n) &= \frac{\theta(-2-\theta(n-p-1)(s+n-p-1)-2n+p+2)}{(n+1)(s+1+n)}, \\ \beta'_2(n) &= \frac{\theta^2(1+n-p-2)}{(n+1)(s+1+n)}.\end{aligned}$$

Proof. It is known to all that:

$$\Gamma(s, z) = e^{-z}U(1-s; 1-s; z), \quad (3.4.1)$$

while $U(a; b; z) = \frac{\Gamma(1-b)}{\Gamma(a+1-b)}M(a; b; z) + \frac{\Gamma(b-1)}{\Gamma(a)}z^{1-b}M(a+1-b; 2-b; z)$ is a combination of two solution of the Kummer's equation.

Now, substitute the the above equation into (3.4.1), we get:

$$\begin{aligned}(1-\theta z)^p \Gamma(s, z) &= e^{-z}(1-\theta z)^p \left(\frac{\Gamma(s)}{\Gamma(1)}M(1-s; 1-s; z) + \frac{\Gamma(-s)}{\Gamma(1-s)}z^s M(1; s+1; z) \right) \\ &= e^{-z} \left(\Gamma(s) \sum_{n=0}^{\infty} u_n z^n + \frac{-1}{s} z^s \sum_{n=0}^{\infty} u'_n z^n \right) = \sum_{n=0}^{\infty} \frac{(-1)^n z^n}{n!} \left(\Gamma(s) \sum_{n=0}^{\infty} u_n z^n + \frac{-1}{s} z^s \sum_{n=0}^{\infty} u'_n z^n \right) \\ &= \sum_{n=0}^{\infty} \Gamma(s) \sum_{k=0}^n \frac{(-1)^k u_{n-k}}{k!} z^n + z^s \sum_{n=0}^{\infty} \frac{-1}{s} \sum_{k=0}^n \frac{(-1)^k u'_{n-k}}{k!} z^n\end{aligned}$$

while the explicit expression of u_n and u'_n could be calculated directly by Theorem 3.2.1, we could just get the final expression:

$$(1-\theta z)^p \Gamma(s, z) = \sum_{n=0}^{\infty} v_n z^n + \sum_{n=0}^{\infty} w_n z^{n+s}$$

with $v_n = \Gamma(s) \sum_{k=0}^n \frac{(-1)^k u_{n-k}}{k!}$, $w_n = \frac{-1}{s} \sum_{k=0}^n \frac{(-1)^k u'_{n-k}}{k!}$, which completes the proof. $\square$

We further state a corollary involving the modified Bessel function $I_{a-\frac{1}{2}}(z)$.

Corollary 3.4.4. *Let $a, \theta, p, z \in \mathbb{C}$, $-2a \notin \mathbb{N} \cup \{0\}$, $|z| < \frac{1}{|\theta|}$ and $|\arg(1-\theta z)| < \pi$, then we have:*

$$(1-\theta z)^p I_{a-\frac{1}{2}}(z) = \sum_{n=0}^{\infty} a_n z^n$$

where

$$\begin{aligned}
a_n &= \frac{4^{\frac{1}{2}-a}}{\Gamma(a + \frac{1}{2})} \sum_{k=0}^n \frac{(-1)^k u_{n-k}}{2^k k!}, \\
u_0 &= 1, u_1 = \frac{1}{2} - \theta p, u_2 = \frac{1}{2} \left(\frac{a+1}{2a+1} - \theta p + \theta^2(p-1)p \right) \\
u_{n+1} &= \beta_0(n)u_n + \beta_1(n)u_{n-1} + \beta_2(n)u_{n-2}, \\
\beta_0(n) &= \frac{a + 2\theta n(2a+n-1) - \theta p(2a+2n) + n}{(n+1)(2a+n)}, \\
\beta_1(n) &= \frac{\theta(-2a - \theta(n-p-1)(2a+n-p-2) - 2n+p+2)}{(n+1)(2a+n)}, \\
\beta_2(n) &= \frac{\theta^2(a+n-p-2)}{(n+1)(2a+n)}.
\end{aligned}$$

Proof. It is known to all that:

$$M(a; 2a; z) = I_{a-\frac{1}{2}}(z) \Gamma(a + \frac{1}{2}) \left(\frac{z}{4}\right)^{\frac{1}{2}-a} e^{\frac{z}{4}}$$

Therefore, applying Theorem 3.2.1 gives

$$\begin{aligned}
(1-\theta z)^p I_{a-\frac{1}{2}}(z) &= e^{-\frac{z}{4}} \frac{4^{\frac{1}{2}-a}}{\Gamma(a + \frac{1}{2})} z^{a-\frac{1}{2}} (1-\theta z)^p M(a; 2a; z) \\
&= \frac{4^{\frac{1}{2}-a}}{\Gamma(a + \frac{1}{2})} z^{a-\frac{1}{2}} \sum_{n=0}^{\infty} \frac{(-1)^n z^n}{2^n n!} \sum_{n=0}^{\infty} u_n z^n \\
&= \frac{4^{\frac{1}{2}-a}}{\Gamma(a + \frac{1}{2})} z^{a-\frac{1}{2}} \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{(-1)^k u_{n-k}}{2^k k!} z^n \\
&= \sum_{n=0}^{\infty} a_n z^{n+a-\frac{1}{2}},
\end{aligned}$$

where u_n could be calculated through Theorem 3.2.1 and $a_n = \frac{4^{\frac{1}{2}-a}}{\Gamma(a+\frac{1}{2})} \sum_{k=0}^n \frac{(-1)^k u_{n-k}}{2^k k!}$, which completes the proof. $\square$

3.4.2 Applications of the Gauss hypergeometric function

First, we present the recurrence relation involving zero-balanced hypergeometric function by substituting $c = a + b$ into Theorem 3.3.1.

Corollary 3.4.5. *Let $a, b, z \in \mathbb{C}$, $-(a+b) \notin \mathbb{N} \cup \{0\}$. Then we have:*

$$(1-\theta z)^p F(a, b; a+b; z) = \sum_{n=0}^{\infty} v_n z^n$$

with $|z| < 1$, $|\arg(1-\theta z)| < \pi$ and $v_0 = 1$, $v_1 = \frac{ab}{a+b} - p\theta$,

$$v_2 = \frac{1}{2} \theta^2 p(p-1) - \theta p \frac{ab}{a+b} + \frac{1}{2} \frac{ab(b+1)(a+1)}{(a+b)(a+b+1)},$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p,\theta}(a,b)}{(n+1)(n+a+b)}v_n - \theta \frac{\zeta_{n,p,\theta}(a,b)}{(n+1)(n+a+b)}v_{n-1} + \theta^2 \frac{\sigma_{n,p}(a,b)}{(n+1)(n+a+b)}v_{n-2},$$

where

$$\begin{aligned}\mu_{n,p,\theta}(a,b) &= (n+a)(n+b) + \theta [2n^2 - 2n(p - (a+b) + 1) - (a+b)p], \\ \zeta_{n,p,\theta}(a,b) &= 2n^2 + 2(a+b-p-2)n - (a+b-1)p + 2(a-1)(b-1) \\ &\quad + \theta(n-p-1)(n-p+a+b-2), \\ \sigma_{n,p}(a,b) &= (n+a-p-2)(n+b-p-2).\end{aligned}$$

Next, we present a recurrence relation involving the product of the logarithmic function and the Gauss hypergeometric function.

Corollary 3.4.6. *Let $a, b, c, z \in \mathbb{C}$, $-c \notin \mathbb{N} \cup \{0\}$. Then we have*

$$V(z) = \ln(1-z) \times F(a, b; c; z) = \sum_{n=0}^{\infty} \lambda_n z^n,$$

with $|z| < 1$, $|\arg(1-z)| < \pi$, $\lambda_0 = 0$, $\lambda_1 = -1$ and for $n \geq 1$,

$$\lambda_{n+1} = \tau_n \lambda_n - \rho_n \lambda_{n-1} + \gamma_n w_n, \quad (3.4.2)$$

where τ_n, ρ_n are defined in (3.3.9), (3.3.10), and

$$\begin{aligned}\gamma_n &= \frac{(c-b-a)n^2 + (a+b-2ab)n - c(a-1)(b-1)}{(n+1)(n+a-1)(n+b-1)(n+c)}, \\ w_n &= \frac{(a)_n(b)_n}{n!(c)_n} = \frac{\Gamma(c)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(n+a)\Gamma(n+b)}{n!\Gamma(n+c)}.\end{aligned}$$

Proof. Notice that

$$F(a, b; c; z) = \sum_{n=0}^{\infty} w_n z^n \quad \text{and} \quad \lim_{p \rightarrow 0} \frac{(1-z)^p - 1}{p} = \ln(1-z),$$

we have

$$V(z) = \ln(1-z) \times F(a, b; c; z) = \lim_{p \rightarrow 0} \frac{(1-z)^p F(a, b; c; z) - F(a, b; c; z)}{p},$$

and then,

$$\sum_{n=0}^{\infty} \lambda_n z^n = \lim_{p \rightarrow 0} \sum_{n=0}^{\infty} \frac{v_n - w_n}{p} z^n = \sum_{n=0}^{\infty} \left(\lim_{p \rightarrow 0} \frac{v_n - w_n}{p} \right) z^n,$$

which implies that

$$\lambda_n = \lim_{p \rightarrow 0} \frac{v_n - w_n}{p} \quad \text{for } n \geq 0.$$

Evidently, $\lambda_0 = 0$, $\lambda_1 = -1$. To obtain (3.4.2), we rewrite the recurrence relation (3.3.8) as

$$\frac{v_{n+1} - w_{n+1}}{p} = \tau_n \frac{v_n - w_n}{p} - \rho_n \frac{v_{n-1} - w_{n-1}}{p} + \frac{\tau_n w_n - \rho_n w_{n-1} - w_{n+1}}{p}.$$

Noting that

$$\begin{aligned}
\frac{\tau_n w_n - \rho_n w_{n-1} - w_{n+1}}{p} &= \frac{w_n}{p} \left[\tau_n - \rho_n \frac{n(n+c-1)}{(n+a-1)(n+b-1)} - \frac{(n+a)(n+b)}{(n+1)(n+c)} \right] \\
&= \frac{(c-b-a-p)n^2 + (a+b-2ab+p-cp)n - c(a-1)(b-1)}{(n+1)(n+a-1)(n+b-1)(n+c)} w_n \\
&\rightarrow \frac{(c-b-a)n^2 + (a+b-2ab)n - c(a-1)(b-1)}{(n+1)(n+a-1)(n+b-1)(n+c)} w_n = \gamma_n w_n,
\end{aligned}$$

the required recurrence relation (3.4.2) just follows as $p \rightarrow 0$, which completes the proof. $\square$

We also obtain a special case corresponding to the complete elliptic integrals of the first and second kinds by substituting the explicit a, b, c into Theorem 3.3.1.

Corollary 3.4.7. *Let $0 < r < 1$, then we have:*

$$\begin{aligned}
(1 - \theta r^2)^p K(r) &= \frac{\pi}{2} (1 - \theta r^2)^p F\left(\frac{1}{2}, \frac{1}{2}; 1; r^2\right) = \sum_{n=0}^{\infty} \frac{\pi}{2} v_n r^{2n} \\
(1 - \theta r^2)^p E(r) &= \frac{\pi}{2} (1 - \theta r^2)^p F\left(-\frac{1}{2}, \frac{1}{2}; 1; r^2\right) = \sum_{n=0}^{\infty} \frac{\pi}{2} v'_n r^{2n},
\end{aligned}$$

with $v_0 = 1$, $v_1 = \frac{1}{4} - p\theta$, $v_2 = \frac{1}{2}\theta^2 p(p-1) - \frac{1}{4}\theta p + \frac{9}{64}$ and $v'_0 = 1$, $v'_1 = -\frac{1}{4} - p\theta$, $v'_2 = \frac{1}{2}\theta^2 p(p-1) + \frac{1}{4}\theta p - \frac{3}{64}$,

and for $n \geq 2$,

$$\begin{aligned}
v_{n+1} &= \frac{\mu(n, p, \theta)}{(n+1)^2} v_n - \theta \frac{\zeta(n, p, \theta)}{(n+1)^2} v_{n-1} + \theta^2 \frac{\sigma(n, p)}{(n+1)^2} v_{n-2}, \\
v'_{n+1} &= \frac{\mu'(n, p, \theta)}{(n+1)^2} v'_n - \theta \frac{\zeta'(n, p, \theta)}{(n+1)^2} v'_{n-1} + \theta^2 \frac{\sigma'(n, p)}{(n+1)^2} v'_{n-2},
\end{aligned}$$

where

$$\begin{aligned}
\mu(n, p, \theta) &= \left(n + \frac{1}{2}\right) \left(n + \frac{1}{2}\right) + \theta [2n^2 - 2np - p], \\
\zeta(n, p, \theta) &= 2n^2 - 2(p+1)n + \frac{1}{2} + \theta(n-p-1)^2, \\
\sigma(n, p) &= \left(n - p - \frac{3}{2}\right)^2,
\end{aligned}$$

$$\begin{aligned}
\mu'(n, p, \theta) &= n^2 - \frac{1}{4} + \theta(2n^2 - 2np - p), \\
\zeta'(n, p, \theta) &= 2n^2 - 2(p+2)n + p - \frac{3}{2} + \theta(n-p-1)^2, \\
\sigma'(n, p) &= \left(n - p - \frac{5}{2}\right) \left(n - p - \frac{3}{2}\right).
\end{aligned}$$

We further provide an explicit power series expansion for the product of a logarithmic function and a power function.

Corollary 3.4.8. *Let $z, p \in \mathbb{C}$, $-p \notin \mathbb{N}$, $|z| < 1$ and $|\arg(1-z)|, |\arg(1+z)| < \pi$, then we have*

$$(1+z)^p \ln(1-z) = - \sum_{n=0}^{\infty} v_n z^{n+1}$$

where

$$v_n = \frac{(-1)^n (1-p)_n}{(n+1)!} \sum_{k=0}^n \frac{(-1)^k k!}{(1-p)_k} \left[\sum_{j=0}^k \frac{(-1)^j (-p-1)_j}{j!} \right]. \quad (3.4.3)$$

If there is a $-p_0 \in \mathbb{N}$ such that $(1-p_0)_k = 0$ for certain $0 \leq k \leq n$, then we define $v_n = \lim_{p \rightarrow p_0} v_n(p)$.

Proof. We only consider the case of $-p \notin \mathbb{N}$. Note that:

$$-\frac{\ln(1-z)}{z} = \sum_{n=0}^{\infty} \frac{z^n}{n+1} = F(1, 1; 2; z).$$

Taking $(a, b, c) = (1, 1, 2)$ in Corollary 3.3.2 gives $v_0 = 1$, $v_1 = \frac{1}{2} + p$, $v_2 = \frac{3p^2+2}{6}$ and for $n \geq 2$,

$$v_{n+1} = -\frac{n-2p-1}{n+2} v_n + \frac{n^2+n-p^2-2p}{(n+1)(n+2)} v_{n-1} + \frac{(n-p-1)^2}{(n+1)(n+2)} v_{n-2}.$$

This 3-order recurrence relation can be reduced to a 2-order one:

$$v'_{n+1} = \frac{p+2}{n+2} v'_n + \frac{n(n-p-1)}{(n+2)(n+1)} v'_{n-1}, \quad n \geq 1. \quad (3.4.4)$$

where

$$v'_n = v_n + \frac{n-p}{n+1} v_{n-1} \quad (3.4.5)$$

with $v'_1 = (p+2)/2$ and $v'_2 = (p^2+3p+4)/6$. Furthermore, the 2-order recurrence relation (3.4.4) can be reduced to 1-order one:

$$v''_{n+1} = -\frac{n-p-1}{n+2} v''_n, \quad n \geq 2, \quad (3.4.6)$$

where

$$v''_n = v'_n - \frac{n}{n+1} v'_{n-1} \text{ with } v''_2 = \frac{1}{6} p(p+1).$$

Solving the recurrence equation (3.4.6) gives

$$v'_n - \frac{n}{n+1} v'_{n-1} = v''_n = \frac{(-1)^n (-1-p)_n}{(n+1)!},$$

which, in combination with (3.4.5), implies that

$$(k+1)v_k + (k-p)v_{k-1} = (k+1)v'_k = \sum_{j=0}^k \frac{(-1)^j (-1-p)_j}{(j+1)!},$$

for $k \geq 1$. Solving this for v_k leads to (3.4.3), which completes the proof. $\square$

The following corollary provides a recurrence relation for the coefficients of the product of $(1-z^2)$ and the inverse hyperbolic tangent function $\operatorname{arctanh} z$.

Corollary 3.4.9. *Let $z \in \mathbb{C}$, $|z| < 1$, then we have:*

$$f(z) = (1 - z^2) \operatorname{arctanh} z = \sum_{n=0}^{\infty} v_n z^{2n+1}$$

with $v_0 = 1$, $v_1 = -\frac{1}{6}$, and for $n \geq 1$,

$$v_n = \tau_n v_n - \rho_n v_{n-1},$$

where

$$\tau_n = \frac{2n^2 - 1}{(n + \frac{3}{2})(n + 1)}$$

$$\rho_n = \frac{(n - 1)(2n - 3)}{(n + 1)(2n + 3)}$$

Proof. It is known to all that we have a Gauss hypergeometric representation for the inverse hyperbolic tangent function:

$$\operatorname{arctanh} z = z F\left(\frac{1}{2}, 1; \frac{3}{2}; z^2\right)$$

Therefore, by Corollary 3.3.3, the multiplication becomes:

$$f(z) = z(1 - z^2) F\left(\frac{1}{2}, 1; \frac{3}{2}; z^2\right) = z \sum_{n=0}^{\infty} v_n z^{2n}.$$

And the coefficient v_n can be calculated through Corollary 3.3.3, which completes the proof. $\square$

The following corollary provides a recurrence relation for the coefficients of Chebyshev polynomials $y = T_m(z) = T_n(\cos \theta) = \cos(n\theta)$, $z \in \mathbb{C}$. Chebyshev polynomials are the solutions to the following equation:

$$(1 - z^2) \frac{d^2 y}{dz^2} - z \frac{dy}{dz} + m^2 y = 0$$

And it has a representation in Gauss hypergeometric function:

$$T_m(z) = F\left(-m, m; \frac{1}{2}; \frac{1 - z}{2}\right) \quad (3.4.7)$$

Corollary 3.4.10. *Let $z, \theta, p \in \mathbb{C}$, $|\frac{1-z}{2}| < 1$ and $|\arg(1 - \theta \frac{1-z}{2})| < \pi$, then we have:*

$$f(z) = (1 - \theta \frac{1-z}{2})^p T_m(z) = \sum_{n=0}^{\infty} v_n \left(\frac{1-z}{2}\right)^n$$

with $v_0 = 1$, $v_1 = -2m^2 - p\theta$,

$$v_2 = \frac{1}{2} \theta^2 p(p - 1) + 2m^2 \theta p + \frac{4m^2(m + 1)(m - 1)}{6},$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p,\theta}(m)}{(n + 1)(n + \frac{1}{2})} v_n - \theta \frac{\zeta_{n,p,\theta}(m)}{(n + 1)(n + \frac{1}{2})} v_{n-1} + \theta^2 \frac{\sigma_{n,p}(m)}{(n + 1)(n + \frac{1}{2})} v_{n-2},$$

where

$$\begin{aligned}\mu_{n,p,\theta}(m) &= (n-m)(n+m) + \theta [2n^2 - 2n(p-c+1) - cp], \\ \zeta_{n,p,\theta}(m) &= 2n^2 - 2(p+2)n + p + 2(-m-1)(m-1) \\ &\quad + \theta(n-p-1)\left(n-p+\frac{1}{2}-2\right), \\ \sigma_{n,p}(m) &= (n-m-p-2)(n+m-p-2).\end{aligned}$$

Proof. We just use equation (3.4.7) to derive:

$$(1 - \theta \frac{1-z}{2})^p T_m(z) = (1 - \theta \frac{1-z}{2})^p F(-m, m; \frac{1}{2}; \frac{1-z}{2}).$$

Then we just apply Theorem 3.3.1 to calculate the explicit expression of recurrence relations, which completes the proof. $\square$

The following corollary provides a recurrence relation for the coefficients of the product of $(1+\theta z)^p$ and $(1+z)^\alpha$.

Corollary 3.4.11. *Let $z, \theta, p, \alpha, \beta \in \mathbb{C}$, $|z| < 1$ and $|\arg(1+\theta z)| < \pi$, then we have:*

$$f(z) = (1+\theta z)^p (1+z)^\alpha = \sum_{n=0}^{\infty} v_n (-z)^n$$

with $v_0 = 1$, $v_1 = \alpha - p\theta$,

$$v_2 = \frac{1}{2}\theta^2 p(p-1) - \theta p\alpha + \alpha(\alpha+1),$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p,\theta}(\alpha, \beta)}{(n+1)(n+\beta)} v_n - \theta \frac{\zeta_{n,p,\theta}(\alpha, \beta)}{(n+1)(n+\beta)} v_{n-1} + \theta^2 \frac{\sigma_{n,p}(\alpha, \beta)}{(n+1)(n+\beta)} v_{n-2}, \quad (3.4.8)$$

where

$$\begin{aligned}\mu_{n,p,\theta}(\alpha, \beta) &= (n+\alpha)(n+\beta) + \theta [2n^2 - 2n(p-\beta+1) - \beta p], \\ \zeta_{n,p,\theta}(\alpha, \beta) &= 2n^2 + 2(\alpha+\beta-p-2)n - (\alpha+\beta-1)p + 2(\alpha-1)(\beta-1) \\ &\quad + \theta(n-p-1)(n-p+\beta-2), \\ \sigma_{n,p}(\alpha, \beta) &= (n+\alpha-p-2)(n+\beta-p-2).\end{aligned}$$

Proof. It is know to all that $(1+z)^\alpha = F(\alpha, \beta; \beta; -z)$, then we substitute it to the above equation and apply Theorem 3.3.1, which completes the proof. $\square$

The following corollary provides a recurrence relation for the coefficients of the product of $(1-\theta z^2)^p$ and $\arcsin z$.

Corollary 3.4.12. *Let $z, \theta, p \in \mathbb{C}$, $|z| < 1$ and $|\arg(1-\theta z^2)| < \pi$, then we have:*

$$f(z) = (1-\theta z^2)^p \arcsin z = \sum_{n=0}^{\infty} v_n z^{2n+1}$$

with $v_0 = 1$, $v_1 = \frac{1}{6} - p\theta$,

$$v_2 = \frac{1}{2}\theta^2 p(p-1) - \frac{1}{6}\theta p + \frac{3}{40},$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p,\theta}}{(n+1)(n+\frac{3}{2})}v_n - \theta \frac{\zeta_{n,p,\theta}}{(n+1)(n+\frac{3}{2})}v_{n-1} + \theta^2 \frac{\sigma_{n,p}}{(n+1)(n+\frac{3}{2})}v_{n-2}, \quad (3.4.9)$$

where

$$\begin{aligned} \mu_{n,p,\theta} &= \left(n + \frac{1}{2}\right) \left(n + \frac{1}{2}\right) + \theta \left[2n^2 - 2n \left(p - \frac{1}{2}\right) - \frac{3}{2}p\right], \\ \zeta_{n,p,\theta} &= 2n^2 - 2(p+1)n + \frac{1}{2} + \theta(n-p-1) \left(n - p - \frac{1}{2}\right), \\ \sigma_{n,p} &= \left(n - p - \frac{1}{2}\right) \left(n - p - \frac{1}{2}\right). \end{aligned}$$

Proof. It is known to all that $\arcsin z = zF(\frac{1}{2}, \frac{1}{2}; \frac{3}{2}; z^2)$, then we substitute it into the above equation and apply Theorem 3.3.1, which completes the proof. $\square$

The following corollary provides a recurrence relation for the coefficients of the product of $(1+\theta z^2)^p$ and $\arctan z$.

Corollary 3.4.13. *Let $z, \theta, p \in \mathbb{C}$, $|z| < 1$ and $|\arg(1 + \theta z^2)| < \pi$, then we have:*

$$f(z) = (1 + \theta z^2)^p \arctan z = \sum_{n=0}^{\infty} (-1)^n v_n z^{2n+1}$$

with $v_0 = 1$, $v_1 = \frac{1}{3} - p\theta$,

$$v_2 = \frac{1}{2}\theta^2 p(p-1) - \frac{1}{3}\theta p + \frac{1}{5},$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p,\theta}}{(n+1)(n+\frac{3}{2})}v_n - \theta \frac{\zeta_{n,p,\theta}}{(n+1)(n+\frac{3}{2})}v_{n-1} + \theta^2 \frac{\sigma_{n,p}}{(n+1)(n+\frac{3}{2})}v_{n-2}, \quad (3.4.10)$$

where

$$\begin{aligned} \mu_{n,p,\theta} &= \left(n + \frac{1}{2}\right) (n+1) + \theta \left[2n^2 - 2n \left(p - \frac{1}{2}\right) - \frac{3}{2}p\right], \\ \zeta_{n,p,\theta} &= 2n^2 - 2 \left(p + \frac{1}{2}\right) n - \frac{1}{2}p + \theta(n-p-1) \left(n - p - \frac{1}{2}\right), \\ \sigma_{n,p} &= \left(n + \frac{1}{2} - p - 2\right) (n - p - 1). \end{aligned}$$

Proof. It is known to all that $\arctan z = zF(\frac{1}{2}, 1; \frac{3}{2}; -z^2)$, then we substitute it into the above equation and apply Theorem 3.3.1, which completes the proof. $\square$

The following corollary provides a recurrence relation for the coefficients of the product of $(1 - \theta \sin^2 z)^p$ and $\cos \xi z$.

Corollary 3.4.14. *Let $z, \xi, \theta, p \in \mathbb{C}$, $|\sin z| < 1$ and $|\arg(1 - \theta \sin^2 z)| < \pi$, then we have:*

$$f(z) = (1 - \theta \sin^2 z)^p \cos \xi z = \sum_{n=0}^{\infty} v_n (\sin z)^{2n}$$

with $v_0 = 1$, $v_1 = -\frac{\xi^2}{2} - p\theta$,

$$v_2 = \frac{1}{2}\theta^2 p(p-1) + \frac{1}{2}p\theta\xi^2 + \frac{\xi^2(\xi^2-4)}{24},$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p,\theta}}{(n+1)(n+\frac{1}{2})}v_n - \theta \frac{\zeta_{n,p,\theta}}{(n+1)(n+\frac{1}{2})}v_{n-1} + \theta^2 \frac{\sigma_{n,p}}{(n+1)(n+\frac{1}{2})}v_{n-2},$$

where

$$\begin{aligned}\mu_{n,p,\theta} &= n^2 - \frac{\xi^2}{4} + \theta \left[2n^2 - (2p+1)n - \frac{p}{2} \right], \\ \zeta_{n,p,\theta} &= 2n^2 - 2(p+2)n + p + 2 - \frac{\xi^2}{2} + \theta(n-p-1) \left(n - p - \frac{3}{2} \right), \\ \sigma_{n,p} &= (n-p-2)^2 - \frac{\xi^2}{4}.\end{aligned}$$

Proof. It is known to all that $\cos \mu z = F(\frac{\xi}{2}, -\frac{\xi}{2}; \frac{1}{2}; \sin^2 z)$, then we substitute it into the above equation and apply Theorem 3.3.1, which completes the proof. $\square$

The following corollary provides a recurrence relation for the coefficients of the product of $(1 - \theta \sin^2 z)^p$ and $\sin \xi z$.

Corollary 3.4.15. *Let $z, \xi, \theta, p \in \mathbb{C}$, $|\sin z| < 1$ and $|\arg(1 - \theta \sin^2 z)| < \pi$, then we have:*

$$f(z) = (1 - \theta \sin^2 z)^p \sin \xi z = \sum_{n=0}^{\infty} v_n (\sin z)^{2n+1}$$

with $v_0 = 1$, $v_1 = \frac{1-\xi^2}{6} - p\theta$,

$$v_2 = \frac{1}{2}\theta^2 p(p-1) + \frac{1}{6}p\theta(\xi^2-1) + \frac{(1-\xi^2)(9-\xi^2)}{120},$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p,\theta}}{(n+1)(n+\frac{3}{2})}v_n - \theta \frac{\zeta_{n,p,\theta}}{(n+1)(n+\frac{3}{2})}v_{n-1} + \theta^2 \frac{\sigma_{n,p}}{(n+1)(n+\frac{3}{2})}v_{n-2}, \quad (3.4.11)$$

where

$$\begin{aligned}\mu_{n,p,\theta} &= n^2 + n + \frac{1-\xi^2}{4} + \theta \left[2n^2 - (2p-1)n - \frac{3p}{2} \right], \\ \zeta_{n,p,\theta} &= 2n^2 - 2(p+1)n + \frac{1-\xi^2}{2} \\ &\quad + \theta(n-p-1) \left(n - p - \frac{1}{2} \right), \\ \sigma_{n,p} &= \left(n - p - \frac{3}{2} \right)^2 - \frac{\xi^2}{4}.\end{aligned}$$

Proof. It is known to all that $\sin \mu z = \mu \sin z F(\frac{1+\xi}{2}, \frac{1-\xi}{2}; \frac{3}{2}; \sin^2 z)$, then we substitute it into the above equation and apply Theorem 3.3.1, which completes the proof. $\square$

The following corollary provides a recurrence relation for the coefficients of the product of $(1 - \theta z^2)^p$

and $\log \frac{1+z}{1-z}$.

Corollary 3.4.16. *Let $z, \theta, p \in \mathbb{C}$, $|z| < 1$ and $|\arg(1 - \theta z^2)|, |\arg(\frac{1+z}{1-z})| < \pi$, then we have:*

$$f(z) = (1 - \theta z^2)^p \log \frac{1+z}{1-z} = \sum_{n=0}^{\infty} 2v_n z^{2n+1}$$

with $v_0 = 1$, $v_1 = \frac{1}{3} - p\theta$,

$$v_2 = \frac{1}{2}\theta^2 p(p-1) - \frac{1}{3}p\theta + \frac{1}{5},$$

and for $n \geq 2$,

$$v_{n+1} = \frac{\mu_{n,p,\theta}}{(n+1)(n+\frac{3}{2})}v_n - \theta \frac{\zeta_{n,p,\theta}}{(n+1)(n+\frac{3}{2})}v_{n-1} + \theta^2 \frac{\sigma_{n,p}}{(n+1)(n+\frac{3}{2})}v_{n-2}, \quad (3.4.12)$$

where

$$\begin{aligned} \mu_{n,p,\theta} &= n^2 + \frac{3}{2}n + \frac{1}{2} + \theta \left[2n^2 - (2p-1)n - \frac{3p}{2} \right], \\ \zeta_{n,p,\theta} &= 2n^2 - (2p+1)n - \frac{p}{2} \\ &\quad + \theta(n-p-1) \left(n - p - \frac{1}{2} \right), \\ \sigma_{n,p} &= (n-p-1) \left(n - p - \frac{3}{2} \right). \end{aligned}$$

Proof. It is known to all that $\log \frac{1+z}{1-z} = 2zF(\frac{1}{2}, 1; \frac{3}{2}; z^2)$, then we just substitute it into the above equation and apply Theorem 3.3.1, which completes the proof. $\square$

Finally, we provide a recurrence relation of coefficients for the product of the inverse trigonometric function and inverse hyperbolic function.

Corollary 3.4.17. *Let $z \in \mathbb{C}$, $|z| < 1$ and $|\arg(1 - z^2)| < \pi$, then we have:*

$$f(z) = \arcsin z \times \operatorname{arctanh} z = \sum_{n=0}^{\infty} v_n z^{2n+2}$$

with $v_n = \frac{v'_n + v''_n}{2n+2}$ and $v'_0 = v''_0 = 1$, $v'_1 = \frac{5}{6}$, $v''_1 = \frac{7}{6}$, where

$$\begin{aligned} v'_{n+1} &= \frac{2n^2 + 3n + \frac{5}{4}}{(n+1)(n+\frac{3}{2})}v'_n - \frac{n(2n+1)}{(n+1)(2n+3)}v'_{n-1} \\ v''_{n+1} &= \frac{2n^2 + \frac{7}{2}n + \frac{7}{4}}{(n+1)(n+\frac{3}{2})}v''_n - \frac{(n+\frac{1}{2})^2}{(n+1)(n+\frac{3}{2})}v''_{n-1} \end{aligned}$$

Proof. For the $f(z) = \arcsin z \times \operatorname{arctanh} z$, we can first differentiate it and get:

$$\begin{aligned} f' &= (1 - z^2)^{-\frac{1}{2}} \operatorname{arctanh} z + (1 - z^2)^{-1} \arcsin z \\ &= \sum_{n=0}^{\infty} v'_n z^{2n+1} + \sum_{n=0}^{\infty} v''_n z^{2n+1}. \end{aligned} \quad (3.4.13)$$

while v'_n and v''_n could be directly calculated through Corollary 3.3.3:

$$\begin{aligned} v'_0 &= v''_0 = 1, v'_1 = \frac{5}{6}, v''_1 = \frac{7}{6} \\ v'_{n+1} &= \frac{2n^2 + 3n + \frac{5}{4}}{(n+1)(n+\frac{3}{2})} v'_n - \frac{n(2n+1)}{(n+1)(2n+3)} v'_{n-1} \\ v''_{n+1} &= \frac{2n^2 + \frac{7}{2}n + \frac{7}{4}}{(n+1)(n+\frac{3}{2})} v''_n - \frac{(n+\frac{1}{2})^2}{(n+1)(n+\frac{3}{2})} v''_{n-1} \end{aligned} \quad (3.4.14)$$

Now, by the Maclaurin expansion of above two special functions, we know that:

$$\begin{aligned} \arcsin z &\sim z + \frac{z^3}{6} + \frac{3}{40}z^5 + o(z^5) \\ \operatorname{arctanh} z &\sim z + \frac{z^3}{3} + \frac{1}{5}z^5 + o(z^5) \end{aligned}$$

Therefore, we can just write the $f(z)$ as a power series and differentiate it:

$$\begin{aligned} f(z) &= \sum_{n=0}^{\infty} v_n z^{2n+2} \\ f'(z) &= \sum_{n=0}^{\infty} (2n+2) v_n z^{2n+1} \end{aligned} \quad (3.4.15)$$

Now compare the coefficients of the z^{2n+1} in (3.4.13) and (3.4.15), we have:

$$v_n = \frac{v'_n + v''_n}{2n+2} \quad (3.4.16)$$

Combine the results of (3.4.14) and (3.4.16), we get the final results, which completes the proof. $\square$

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